Fault-Tolerant Magic-State Cultivation: An Exact Criterion and Checks That Keep the Distance

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Magic-state cultivation prepares T states with physical T gates, noisy Clifford gates and postselection. Replacing each T gate by an injected T state, we show that a fault configuration is harmless exactly when it triggers a detector of the remaining Clifford circuit or one stabilizer code projects the perturbed T states onto the encoded target or to zero. The code must distill the target to the order of the fault distance d, and every such code is attained: cultivation needs exactly as many T gates as distillation needs inputs, at least $3d - 6$. Because the checks of cultivation are deterministic, the criterion reduces to two exponential sums of $\mathbb{Z}_4$-valued quadratic forms and certifies fault distances with real T gates in polynomial time. GHZ-fold checks as in current protocols, including the circuit of Google's cultivation experiment, have fault distance $(d + 1)/2$ at code distances three to seven, while verified-cat checks keep the full distance. On a square grid with nearest-neighbour gates they keep it at distances three and five with 17 and 46 qubits, accept more often than any fold, and at distance three fail 10^3 times less often than the checks of the experiment.

INTRODUCTION

Magic-state cultivation [1] prepares a T state by injecting it into a small color code, checking it with circuits built from physical T and $T^\dagger$ gates, and postselecting on every check, before the code is grown into a surface code. It has been demonstrated on hardware [2], and in simulations it reaches logical error rates that previously required distillation, at a cost comparable to that of a lattice-surgery CNOT [1]. Variants use other codes and layouts [3, 4], and related schemes inject a state and verify it by postselection [5–8]. These circuits are designed by hand and assessed by simulation, originally with S gates in place of the non-Clifford T gates. Exact simulators now keep the T gates [9–14], and with them the distance-three and distance-five circuits of Ref. [1] turn out to have fault distances two and three, not three and five [15, 16]. What is missing is a criterion for fault tolerance that applies to every such circuit and yields lower bounds; we know of no lower bounds on the cost of cultivation.

For distillation, the corresponding questions have exact answers in small cases [17]: Clifford operations are ideal, only the T-state inputs are noisy, and every protocol, however adaptive, is equivalent for fault tolerance to the projection of its inputs onto a single stabilizer code. Cultivation differs in two respects: every operation is noisy, and a Pauli fault before a T gate becomes a non-Pauli error after it. We show that both differences can be absorbed into a characterization of the same kind. For each accepted run, the tolerance to circuit-level Pauli noise is decided by one stabilizer code on the T gates of that run and a linear map from faults to Pauli errors on those T gates. The code must distill the target to the order of the fault distance, it is then a quantum code of at least that distance, and every such code is attained by some circuit. Hence fault distance d needs exactly as many T gates as distillation of order d needs noisy inputs, for every target, and coding bounds raise the lower bounds for both to about $3d$. For a given circuit, the fault distance lies between the order of its code and the least number of faults that the Clifford part spreads to a heavier error on the T gates. For the distance-three circuit of Ref. [1] these are three and two, and we recover the fault distance two and the leading error coefficient of Refs. [15, 16].

Because every check of a cultivation circuit is deterministic on the target, the criterion also has an exact closed form. A Z error on the T gates is then a stabilizer of the code, its logical Z, or rejected with certainty, and the output of every fault configuration is a combination of the target and its logical error whose two coefficients are exponential sums of $\mathbb{Z}_4$-valued quadratic forms (Theorem 5). A configuration is harmful exactly when the second sum is nonzero, which is decided in polynomial time, so fault distances with real T gates can be certified as fast as for Clifford circuits; we certify them up to code distance seven. A lemma on coherent errors (Lemma 12) gives the mechanism of the loss: an undetected X error on part of a logical operator that reaches the last T layer is completed by further faults into a logical error, while the S proxy detects it. GHZ-fold checks built like those of Ref. [1] have fault distance two, three and four at code distance three, five and seven, so the loss does not saturate (Proposition 14), and the circuit of Google's experiment [2] is one of them (Proposition 18). Checks in which every data qubit interacts once with a verified cat state keep the full distance (Proposition 16). On a square grid, a cat grown from the middle of a path and verified at its ends keeps it with nearest-neighbour gates, and it is the cheapest check we found under superconducting noise (Proposition 17).

SETTING

Circuits and protocols. Let $|T\rangle = (|0\rangle+\omega|1\rangle)/\sqrt{2}$ with $\omega = e^{i\pi/4}$. A *Clifford+T protocol* acts on qubits initialized in $|0\rangle$ with Clifford gates, T and $\mathrm{T}^\dagger$ gates, Pauli measurements, classical randomness, feedforward of any operation on earlier outcomes, postselection and discarding, and outputs r qubits. Its target is a pure state τ with stabilizer nullity $\nu(\tau) = r$, that is, fixed by no Pauli operator other than the identity, as $|T\rangle$ and $|\mathrm{CCZ}\rangle$ are. A protocol is *non-adaptive and fully postselected* if it is a fixed sequence of Clifford gates, T and $\mathrm{T}^\dagger$ gates, projections $(I + sQ)/2$ with a Pauli Q and a fixed sign s, and removals $\langle k|_q$ of qubits with fixed $k \in \{0, 1\}$; $\mathrm{T}^\dagger$ gates count as T gates. Such a circuit has a single Kraus operator K, and we call it *correct* if $K|0\rangle$ is a nonzero multiple of $|\tau\rangle$.

Faults. Every preparation, gate and measurement is a *location*, except those declared ideal; removals are ideal. A fault at a location is a Pauli operator on its support, applied immediately after a preparation or gate or immediately before a projection, or the flip of a projection's sign. Faults at different locations compose, and the *fault space* $\mathbb{V}$ is the $\mathbb{F}_2$-vector space of all assignments of Pauli operators and flips to locations, with the weight $\mathrm{wt}(F)$ of $F \in \mathbb{V}$ the number of locations at which F acts. Each location suffers each of its faults independently with probability $p\lambda > 0$, so a run of an adaptive protocol is described by its random choices and observed outcomes b together with the faults F at the locations executed along b, and has Kraus operator $K_{b,F}$. The pair (b, F) is *harmless* if $K_{b,F}|0\rangle \in \mathbb{C}|\tau\rangle$, which includes rejection. The output error after postselection is $\epsilon = O(p^d)$ exactly when every run with $\mathrm{wt}(F) < d$ is harmless and some fault-free run is accepted; the largest such d is the *fault distance*. It depends only on which faults are possible, not on their probabilities as long as these are of order p, so it is the same, for example, under depolarizing noise, which applies one random Pauli operator per location.

Ideal decoders. A protocol whose output is encoded is judged after an ideal decoder: projections onto the stabilizers of an $[[m, r, \delta]]$ code Q on m output qubits, followed by an ideal Clifford unencoding. We call the decoder *admissible* if the fault-free state at its input lies in Q, so that it accepts the fault-free run with certainty. Its projections and the unencoding are ideal operations of the circuit, so the results below apply to the circuit together with its decoder. Simulations of cultivation use decoders of this kind [1]. Without the condition, the decoder could carry out the T-dependent checks itself: nine noisy T states handed to an ideal decoder of the code of Proposition 10 below would already tolerate every single fault.

ONE CIRCUIT, ONE CODE

Proposition 1. *Let a Clifford+T protocol have fault distance d, and let b be any accepted fault-free run. The non-adaptive, fully postselected circuit formed by the operations executed along b, with every measurement replaced by the projection onto its outcome in b and every discarded qubit removed with an outcome of nonzero amplitude, is correct and has fault distance at least d.*

The proof (Sec. S1) uses only that a sum of positive operators is a multiple of a pure state exactly when each term is. Every lower bound on the qubits, gates or T gates of correct non-adaptive circuits of fault distance d therefore holds for every accepted fault-free run of every protocol of fault distance d.

Theorem 2. *Let a non-adaptive, fully postselected Clifford+T circuit with t T gates be correct for a target τ with $\nu(\tau) = r$. There are a stabilizer code C on t qubits of dimension 2^r with $\Pi_C|T\rangle^{\otimes t} \neq 0$, a Clifford isometry V with $V\Pi_C|T\rangle^{\otimes t} \propto |\tau\rangle$, and $\mathbb{F}_2$-linear maps D, from $\mathbb{V}$ to binary vectors with one entry per projection, and $E\colon \mathbb{V} \to \mathcal{P}_t/\langle iI, S_C\rangle$, into the Pauli group on t qubits modulo phases and the stabilizers of C, such that a fault configuration F is harmless if and only if $D(F) \neq 0$ or*

$$\Pi_C\, E(F)\, |T\rangle^{\otimes t} \in \mathbb{C}\, \Pi_C |T\rangle^{\otimes t}. \tag{1}$$

A Z fault immediately before or after the jth T gate has $D = 0$ and $E = Z_j$.

The construction behind the theorem is the injection gadget [18]: for every single-qubit state,

$$\sqrt{2}\, \langle 0|_a\, \mathrm{CX}_{q\to a}\, |\psi\rangle_q |T\rangle_a = T|\psi\rangle_q, \tag{2}$$

and $\mathrm{T}^\dagger = S^\dagger T$. Replacing every T gate by Eq. (2), with its own ideal ancilla in $|T\rangle$, leaves a noisy Clifford circuit, the *skeleton*, acting on t ideal T states. Its fault-free Kraus operator is a code projection followed by a Clifford isometry, $c\, V\Pi_C$, as in Ref. [17]. A Pauli fault anywhere in the skeleton can be pushed to the end of the circuit, where it flips the signs of the projections it anticommutes with and leaves a Pauli operator on the output. Sign flips that contradict a deterministic outcome annihilate the circuit; these are the detectors, $D(F) \neq 0$. All other sign flips, and the output Pauli, are equivalent to a Pauli on the T inputs, by the Gottesman–Knill description of stabilizer circuits [19]. This gives $E(F)$, and Eq. (1) is the condition that the perturbed T states still project onto the encoded target, or onto zero (Sec. S1).

The pair (D, E) plays the role of the detector error model of a Clifford circuit [20–22], with the logical observables replaced by the Pauli group on the T gates and the test on them by the collapse condition (1), which is the one that decides distillation [17]. Both maps are computed in polynomial time by following the Choi state of the skeleton symbolically; only the collapse condition involves the T states (Sec. S4).

Detectors and checks. The entries of D belong to the projections whose outcome is fixed in the skeleton. A projection whose outcome is fixed in the circuit but depends on the T inputs in the skeleton, such as a check after a layer of T gates, is not a detector in this sense. It is part of the code C, and it rejects a fault only through Eq. (1), that is, only the components of $E(F)|T\rangle^{\otimes t}$ that leave the code. In the distance-three circuit of Ref. [1], 4 of the 20 projections are of this kind: the X measurement that reads out the logical check and three X-type stabilizers measured after the final T layer. Wan and Zapirain likewise find that 16 of its 20 detectors remain Pauli webs with T gates [16]; with S gates the circuit is Clifford, and all 20 are detectors.

Related descriptions. Replacing T gates by gadgets is standard in simulation [23]. For cultivation, Surti *et al.* observed that every Pauli fault reaches the end of the circuit as a Clifford error [14], Chan *et al.* computed the acceptance and fidelity of a code state under such an error [15], and exact simulators follow faults through the T gates [9–13, 16, 24]. Theorem 2 instead moves the non-Clifford part to the inputs, where one code, independent of the fault, judges a Pauli operator that depends linearly on the fault. The collapse condition is itself combinatorial: it compares signed weight distributions of the cosets of S_C in its normalizer (Proposition 22, Sec. S2).

Coherent faults. The new feature relative to distillation is that $E(F)$ may have an X part. Since $X|T\rangle = (|T\rangle + iZ|T\rangle)/\sqrt{2}$ and $XZ|T\rangle = -(i|T\rangle + Z|T\rangle)/\sqrt{2}$, a Pauli $E = X^u Z^v$ turns $|T\rangle^{\otimes t}$ into a superposition, with amplitudes of modulus $2^{-|u|/2}$, of the states $Z^{v+w}|T\rangle^{\otimes t}$ with w supported on u. For a Pauli operator E on the T inputs, let $|E|_S$ be the least number of qubits on which an element of $E\langle iI, S_C\rangle$ acts.

Lemma 3. *(i) If every pattern $v + w$ with $\operatorname{supp} w \subseteq \operatorname{supp} u$ satisfies the distillation condition $\Pi_C Z^{v+w}|T\rangle^{\otimes t} \in \mathbb{C}\,\Pi_C|T\rangle^{\otimes t}$, then so does $X^u Z^v$, and every F with $E(F) = X^u Z^v$ is harmless. (ii) If every Z pattern of weight below d satisfies it, then so does every Pauli operator E with $|E|_S < d$, and C has distance at least d.*

The smallest instance of (i) is a single check (Fig. 1a): a projection onto $|T\rangle$ rejects $Z|T\rangle$ and returns $X|T\rangle$ and $Y|T\rangle$ to $|T\rangle$ half of the time, so X and Y faults before it are harmless although they are neither detected nor stabilizers (Sec. S3). The converse of (i) can fail, because harmful components may cancel. Part (ii) is where cultivation and distillation meet: a code that tolerates Z faults on fewer than d inputs tolerates every Pauli fault on them, and, since no Pauli operator fixes the target, it has no logical operator of weight below d.

Deterministic checks. Lemma 3(i) decides harmlessness when every component collapses. In cultivation, the components behave as on a stabilizer code, and the collapse condition can be decided exactly, without enumerating the cosets of S_C as in Proposition 22.

Definition 4. A correct circuit for $|T\rangle$ has *deterministic checks* if there are a subspace $\Sigma \subseteq \mathbb{F}_2^t$ and $\ell \in \mathbb{F}_2^t$ such that $Z^e \in \pm S_C$ for $e \in \Sigma$, Z^ℓ commutes with S_C and $VZ^\ell V^\dagger$ acts on the output as Z, and $\Pi_C Z^e|T\rangle^{\otimes t} = 0$ for every $e \notin \Sigma \cup (\ell + \Sigma)$.

Every Z error on the T gates is then a stabilizer of C, the logical Z, or rejected with certainty. This is the case when every check measures an operator of which the target is an eigenstate, and not for code-based distillation, where Z errors on the inputs are detected only with some probability. The signs $Z^e = \sigma(e) g_e$, $g_e \in S_C$, define a character σ of Σ, which we extend to a character χ of $\mathbb{F}_2^t$.

Theorem 5. *For a circuit with deterministic checks and a configuration with $D(F) = 0$ and $E(F) = X^a Z^b$,*

$$\Pi_C E(F)|T\rangle^{\otimes t} = \mu\, 2^{-|a|/2}\big(g_0 + g_1 Z^\ell\big)\Pi_C|T\rangle^{\otimes t}, \qquad (3)$$

with $|\mu| = 1$ and, for $c \in \{0, 1\}$,

$$g_c = \sum_{w \le a,\ w+b \in c\ell+\Sigma} i^{|w|}\, \chi(w + b + c\ell). \qquad (4)$$

The configuration is accepted with probability $2^{-|a|}(|g_0|^2 + |g_1|^2)$ relative to the fault-free run, its wrong output has weight $2^{-|a|}|g_1|^2$, and it is harmful if and only if $g_1 \neq 0$.

Definition 4 removes every Z pattern outside the two cosets, and the two surviving terms are orthogonal and of equal norm because $\langle T|Z|T\rangle = 0$ (Sec. S1). The sums (4) run over affine subspaces $w_0 + \langle k_1, \dots, k_m\rangle$ of $\mathbb{F}_2^{|a|}$, and by inclusion and exclusion $|w_0 + \sum_l z_l k_l| \equiv |w_0| + \sum_l z_l(|k_l| - 2|w_0 \wedge k_l|) - 2\sum_{l<m} z_l z_m |k_l \wedge k_m| \pmod 4$, so each g_c is an exponential sum $\sum_{z \in \mathbb{F}_2^m} i^{q(z)}$ of a $\mathbb{Z}_4$-valued quadratic form.

Proposition 6. *Each g_c is computed with $O(t^3)$ operations, and $|g_c|$ is zero or a power of $\sqrt{2}$. In particular the acceptance and the bad weight of every fault configuration are dyadic rationals.*

Summing out one variable of q multiplies the sum by $1 \pm i$ and adds a parity to q, or multiplies it by two and eliminates a second variable through a linear constraint, or shows that it vanishes (Sec. S1). The coherence of real T gates enters only through these sums. Treating the terms of Eq. (4) one at a time, and calling a configuration harmful when one of them lies in $\ell + \Sigma$, gives a lower bound on the fault distance that equals one for every circuit we studied: a single X fault before a check has such a term, which the check removes only through the cancellation $g_1 = 0$.

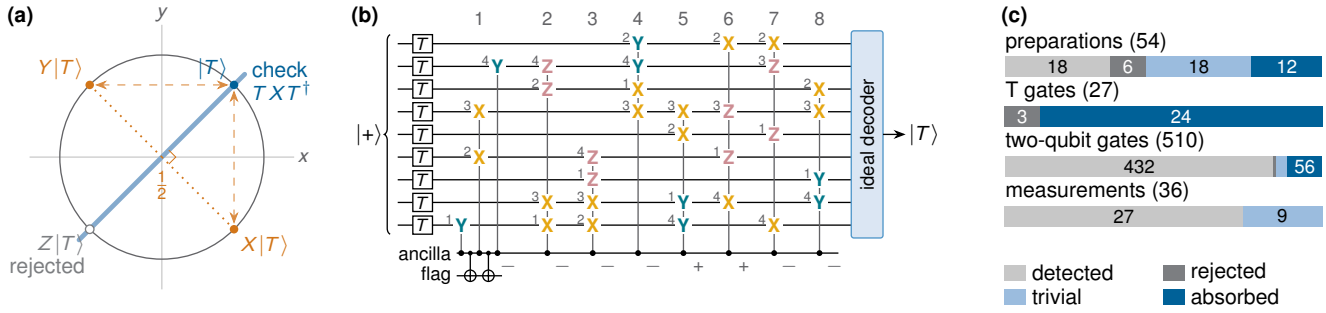


FIG. 1. Fault distance two with nine T gates. (a) A check that projects onto $|T\rangle$, drawn on the equator of the Bloch sphere. A Z fault maps $|T\rangle$ to the antipodal, orthogonal state $Z|T\rangle$, which the check rejects. X and Y faults reflect $|T\rangle$ in the x and y axes; the reflected states lie at right angles to the check axis, so the check accepts them with probability $1/2$ and then outputs exactly $|T\rangle$. Faults of this kind are harmless although no detector sees them. (b) The circuit of Proposition 10. Nine qubits are prepared in $|+\rangle$ and receive one T gate each. Eight stabilizers of the nine-input factory code are then measured in turn (columns 1 to 8): each column couples the ancilla by controlled-Pauli gates to the four marked qubits, with the marked Pauli operators, in the order given by the small numbers, and keeps the indicated outcome. The first measurement carries a flag, coupled to the ancilla after the first and before the last of these gates, which detects errors of the ancilla that would spread to two data qubits. The output remains encoded in the factory code, which has distance three, and an ideal decoder measures its stabilizers and unencodes. (c) The 627 single faults of the circuit, by location type. A fault is detected (a flag, or a stabilizer measured both before and after it, changes its outcome); rejected (its Pauli on the T inputs projects them to zero); trivial (that Pauli is a stabilizer of the code on the T gates); or absorbed (the projection is a nonzero multiple of the encoded T state, as in panel a). The narrow segments of the two-qubit gates are 4 rejected and 18 trivial faults. No single fault is harmful.

Exact evaluation. Theorems 2 and 5 are constructive, and we use two independent implementations. The first follows the Choi state of the skeleton symbolically and decides the collapse condition by state vectors (Sec. S4); it serves for the small circuits. The second takes each ancilla a_j of Eq. (2) as half of an ideal Bell pair with a reference qubit r_j and decodes the data block ideally onto one qubit at the end. The skeleton is then a Clifford circuit whose fault-free output is the Choi state $|\Phi\rangle$ of $V\Pi_C$, a stabilizer state on the t references and the output. One stim detector error model of the skeleton lists every fault class with its probability, its detector pattern D and its error E, read from the flips of the stabilizers of $|\Phi\rangle$. Writing $|\Phi\rangle$ as a sum over an affine subspace of dimension k, one Walsh–Hadamard transform of length 2^k evaluates $\Pi_C Z^e|T\rangle^{\otimes t}$ for all Z patterns e at once; it finds Σ and ℓ and checks Definition 4. Every further evaluation is a Gauss sum, whose cost does not depend on k. The fault distance follows from a meet-in-the-middle search on detector patterns, and the logical error from sampling the error model with the exact weight of each sample (Sec. S5).

We validated the method on the $d = 3$ circuit of Ref. [1] (15 qubits, $t = 15$, $k = 5$) against a brute-force state-vector simulation with real T gates: for all 2,444 single physical faults and 400 random pairs, rejection agrees, and acceptance and bad weight agree to 3×10^{-16}; so do all 96 malignant pairs, which are the 96 harmful pairs that the first implementation finds under the same noise (Sec. S7), with the same leading coefficient $32/75$. The Gauss sums agree with a direct contraction of $|\Phi\rangle$ with $|T\rangle^{\otimes t}$ to 5×10^{-16} on 1,131 random fault classes with k from 5 to 20, and at $k = 20$ they take 1 ms per class instead of 0.22 s. With the S gate in place of T, sampling agrees with direct stim sampling of the proxy circuit. Every circuit we study has deterministic checks. Certifying fault distance five takes 22 s for a distance-five check, and the weight-four certificates at distance seven take five minutes on one workstation.

CULTIVATION AS DISTILLATION

Corollary 7. *In a noise model in which every T gate may suffer a Z fault, every accepted run of a protocol of fault distance d executes at least $n_{\min}(\tau, d)$ T gates, the least number of noisy T inputs of any order-d distillation protocol for τ.*

Proof. By Proposition 1 it suffices to consider a correct non-adaptive circuit with t T gates and fault distance d. Configurations of Z faults at T gates have $D = 0$ and $E = Z^e$, so by Theorem 2 each with $|e| < d$ satisfies $\Pi_C Z^e|T\rangle^{\otimes t} \in \mathbb{C}\,\Pi_C|T\rangle^{\otimes t}$. Measuring the stabilizers of C on t noisy T states, postselecting and decoding with V is therefore an order-d distillation protocol [17]. □

The proof uses only the *order* d_C of the code, the least weight of a Z pattern that violates the distillation condition, which is the suppression order of the distillation protocol that measures C on noisy T states. The Clifford part limits the fault distance in a second way, by spreading faults. We call $d_{\mathrm{sp}} = \min\{\mathrm{wt}(F) : D(F) = 0,\ |E(F)|_S > \mathrm{wt}(F)\}$, the least number of undetected

faults whose Pauli operator on the T inputs is heavier than their number, the *spreading distance* of the circuit.

Theorem 8. *A correct non-adaptive circuit with code C has fault distance d with* $\min(d_C, d_{\rm sp}) \le d \le d_C$. *Conversely, for every stabilizer code C on t qubits that encodes a target as $\Pi_C|T\rangle^{\otimes t} \neq 0$, there is a correct non-adaptive circuit with t T gates and an admissible ideal decoder whose code is C and whose fault distance is d_C.*

The upper bound is the proof of Corollary 7, and the lower bound is Lemma 3(ii). For the converse (Sec. S1), the circuit applies T to t qubits in $|+\rangle$, so that its T inputs are its data, and measures the generators of C one after another with cat states [25], each checked by $d_C - 1$ rounds of parity measurements. A cat state that passes a fault-free round has no X error, so k faults in one measurement leave a data error of weight at most k; the output stays encoded in C, and the ideal decoder, which measures C again, rejects every run whose recorded signs disagree with the data error. Hence $d_{\rm sp} \ge d_C$.

Corollary 9. *Let $\nu(\tau) = r$. The least number of T gates in an accepted run of a protocol of fault distance at least d is $n_{\min}(\tau, d)$. It is at least the least length of a stabilizer code with r logical qubits and distance d, and at least $3d - 4$ for odd d and $3d - 6$ for even d.*

The equality combines Corollary 7 with Theorem 8 for the code of an optimal distillation protocol, which has distance at least d by Lemma 3(ii); the last bounds follow from Rains' theorem that a quantum code of length n corrects at most $\lfloor (n+1)/6 \rfloor$ errors [26]. The exact values of Ref. [17] thus hold for cultivation: fault distance two needs nine T gates for one T state and eight for $|\mathrm{CCZ}\rangle$. The distance bound is new for distillation too. The linear-programming bound for stabilizer codes with Rains' shadow enumerator [26, 27], certified in exact arithmetic (Table SI), raises the lower bounds for one T state at $d = 6, \ldots, 12$ from $\max(2d, d+7)$ to 16, 17, 22, 23, 28, 29 and 34, and for $|\mathrm{CCZ}\rangle$ at $d = 4, \ldots, 8$ from 11, 12, 13, 15 and 17 to 12, 14, 17, 20 and 23.

THE SMALLEST CASE

By Corollary 9, fault distance two for one T state needs exactly nine T gates. The construction of Theorem 8 with the nine-input factory code of Ref. [17] uses 16 qubits and 128 two-qubit gates; an integer program over the fault map finds a smaller circuit.

Proposition 10. *There is a circuit with nine T gates, eleven qubits and 34 two-qubit gates, ending in an admissible ideal decoder, that prepares one T state with fault distance two.*

The circuit (Fig. 1b) prepares nine qubits in $|+\rangle$ and applies T to each, so that its T inputs are its data qubits.

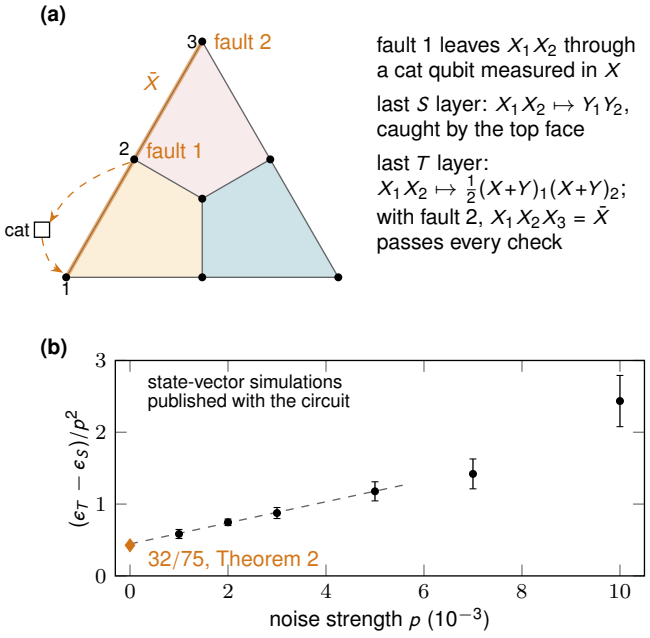


FIG. 2. The distance-three cultivation circuit of Ref. [1] with T gates. (a) One of its 48 harmful pairs of faults, on the seven data qubits of the color code (faces shaded), with the mechanism of Ref. [15]. Fault 1, an X error on qubit 2, is copied to qubit 1 through a qubit of the cat state, which is measured in the X basis. After the last layer of S gates of the stabilizer simulation the errors become Y_1Y_2, which the top face detects; after the last layer of T gates the component X_1X_2 survives, and fault 2 on qubit 3 completes the logical operator $\bar{X}$ along the left edge. (b) Difference of the logical errors per accepted run with T and with S gates, divided by p^2, from the state-vector simulations published with the circuit [28] (one standard deviation; dashed: weighted linear fit for $p \le 5 \times 10^{-3}$), and its limit 32/75 at $p \to 0$ from Theorem 2, which is also the exact leading coefficient of Ref. [16].

It then measures eight independent stabilizers of weight four of the nine-input factory code of Ref. [17], each with one ancilla and four controlled-Pauli gates, and postselects on the outcomes of the factory, which occur with probability 1/128 without faults. That code has distance three, so the output stays encoded in it, and the decoder, which re-measures the stabilizers, is admissible. Faults on the preparations and T gates are single-qubit Paulis on the inputs, which collapse because the code distills to order two, by Lemma 3(ii). A fault while the kth stabilizer is measured leaves a Pauli R on the data and may flip the outcome. It is detected if R anticommutes with a stabilizer measured before, or if the flip does not match the commutation of R with the kth stabilizer; otherwise it acts as R on the inputs, and every Pauli of weight two on the inputs is harmful (Sec. S6). The weight-two errors that the measurement circuits spread to the data must therefore anticommute with a stabilizer measured earlier, or be caught by a flag [29]. Choosing the eight stabilizers, their order, the order of the qubits in each

measurement and the flags is an integer program over the fault map (Sec. S6). The first measurement needs a flag, since no stabilizer precedes it, and Gurobi shows in under a minute that one flag suffices. The proof is exact: the remaining single faults reduce to Paulis on the inputs of weight at most one modulo the stabilizers. An enumeration of all 627 single faults confirms it, with the fault map and by direct simulation (Fig. 1c).

The construction is a distillation at the physical level, in the spirit of Ref. [8], made fault tolerant by the order of its measurements. Idle faults do not lower its fault distance, since an idle fault equals a fault after the preceding operation on the same qubit. It minimizes T gates, not cost: it accepts one fault-free run in 128, and under the uniform depolarizing noise of Ref. [1] its logical error per accepted run is $53p^2$ to leading order without idle noise, and $1.9 \times 10^3 p^2$ with idle noise in an as-soon-as-possible schedule (Sec. S6). The distance-three circuit of the next section has the same fault distance with 15 T gates, 15 qubits and 81 CNOTs, accepts every fault-free run and has error $32p^2/75$.

For fault distance three, Theorem 8 applied to the 15-to-1 code of Bravyi and Kitaev [30] gives a circuit with 15 T gates. Enumerating all pairs of its faults confirms the fault distance, and shows that one round of checks per cat state already suffices here, which leaves 24 qubits and 310 two-qubit gates (Sec. S2). The least number of T gates for fault distance three is therefore between ten and fifteen.

A PUBLISHED CIRCUIT

Stabilizer simulations of cultivation replace T by S, which turns the circuit into a Clifford circuit [1]. With T gates, the distance-three circuit of Ref. [1] has fault distance two [15, 16]. Theorem 2 recovers this, with every class of faults decided exactly.

Proposition 11. *With T gates at the positions where its stabilizer simulations use S gates, the distance-three cultivation circuit of Ref. [1], unitary injection followed by one T check, has fault distance two. With S gates, no pair of faults is harmful.*

We compose the circuit from the published chunks [28] (15 qubits, 15 T gates, 81 CNOTs) and keep the final stabilizer measurements as an admissible ideal decoder, as in Ref. [1]. With every operation a noisy location, no single fault is harmful, and 48 of the 966,391 pairs are. Fig. 2a shows one of them, with the mechanism identified in Ref. [15]: the last layer of T gates turns an X error on two qubits into $[(X+Y)/\sqrt{2}]^{\otimes 2}$, and the final X-type checks, which act only through C, remove the components with Z parts but not $X \otimes X$. Direct simulation confirms every harmful pair. Theorem 8 separates the two causes. The code on the 15 T gates has order $d_C = 3$, with 35 harmful Z patterns of weight three, as many as the 15-to-1 code, so a circuit with the same code reaches fault distance three. The published circuit has spreading distance two: each harmful pair leaves a Pauli operator of weight five modulo S_C on the T inputs, and no undetected pair whose Pauli is lighter is harmful (Sec. S2).

Under the uniform depolarizing noise of Ref. [1], with idle noise in every time step of the chunks, 96 pairs are harmful, and the logical error per accepted run is $\epsilon_T = 32p^2/75 + O(p^3)$, while $\epsilon_S = O(p^3)$ (Sec. S7). The coefficient equals the exact one of Wan and Zapirain [16], obtained by tensor-network contraction of an independently compiled circuit. The state-vector simulations published with the circuit [28], extrapolated linearly to $p = 0$, give $(\epsilon_T - \epsilon_S)/p^2 = 0.45 \pm 0.08$ (Fig. 2b). At $p = 10^{-3}$ the measured difference, $(5.8 \pm 0.6) \times 10^{-7}$, is more than half of ϵ_T, and at smaller p it dominates. Chan *et al.* restore fault distance three with two flag qubits that add four layers of CNOTs [15]. Flags that each compare the Z value of one qubit at two times do not suffice: a set-cover program over the fault map shows that they miss two of the 48 pairs (Sec. S7).

The distance-five circuit. Two properties of the $d = 5$ circuit of Ref. [1] only appear with real T gates. On the 19-qubit color code, the transversal check $\bigotimes_q U_q X U_q^\dagger$, $U_q \in \{T, T^\dagger\}$, is logical only if every face f contains $|f|/2 \bmod 2$ $\mathrm{T}^\dagger$ gates [15], and, among such patterns, the parity of the total number of $\mathrm{T}^\dagger$ gates decides whether the check stabilizes the injected state $T^\dagger|+\rangle$ or its conjugate; with the wrong parity the check projects instead of detecting, half of the runs are lost, Definition 4 fails and the circuit has fault distance two. The same holds at $d = 3$, where transversal T acts on the $[[7, 1, 3]]$ code as the logical $\mathrm{T}^\dagger$. Random outcomes during the growth from $d = 3$ to $d = 5$ leave X-type Pauli frames, which with real T gates must be undone before the $d = 5$ T layers, as in Ref. [9]; we analyse the corresponding branch of maximal weight. With these choices, the full protocol has fault distance three, with leading term $1.5\,p^3$ under uniform noise (Table I), extending the analysis of the final double check in Refs. [15, 16], and its logical error agrees with the GPU simulation of Ref. [9] (Sec. S7).

WHY REAL T GATES LOSE FAULT DISTANCE

Lemma 12 (Coherent completion)**.** *Let a correct circuit apply a transversal layer $U = \bigotimes_q U_q$, $U_q \in \{T, T^\dagger\}$, to the data of a CSS code of distance δ and then decode ideally, and let ℓ be the support of a logical X operator of weight δ. Suppose a fault configuration leaves, immediately before U, the Pauli X^u on the data with $\emptyset \neq u \subsetneq \ell$, that further faults apply X on $\ell \setminus u$ after U, and that no detector is triggered. If no Z stabilizer other than the identity is supported in u, the configuration is accepted*

with probability $2^{-|u|}$ *relative to the fault-free run and outputs* $X_L|\tau\rangle$*, which is harmful for a T state. In the S proxy the same faults are detected.*

Since $U_q X U_q^\dagger = \bar{\omega}^{\pm1} S^{\pm1} X$ and $S^{\pm1} \propto (I \mp iZ)/\sqrt{2}$, the error after the layer is a superposition $2^{-|u|/2} \sum_{w \subseteq u} c_w Z^w X^\ell$ with $|c_w| = 1$. The decoder keeps only the components with trivial syndrome; as $|w| < \delta$ and no Z stabilizer lies in u, only $w = \emptyset$ survives. In the proxy, $SXS^\dagger$ is a Pauli, so the error is $Z^u X^\ell$, whose Z part has a nontrivial syndrome (Sec. S1). A Y in place of the last X gives a fully wrong output.

Corollary 13. *The fault distance of such a circuit is at most* $s + \delta - |u|$ *for every configuration of* s *faults that leaves an undetected* X^u *as in Lemma 12 immediately before the last T layer; in particular it is at most* $1 + h$*, where* h *is the least number of faults that leave such an* X^u *with* $|u| = \delta - 1$*.*

A check that keeps fault distance δ with real T gates must therefore be fault tolerant *relative to the code*: no s undetected faults may place X on more than s qubits of a minimum-weight logical support before the last T layer. The proxy imposes no such condition, because the S layer turns these errors into Pauli errors with a detectable Z part. The condition is much weaker than ordinary fault tolerance of the check ancilla. At $d = 3$ every pair of data qubits lies in a minimum-weight logical support, but at $d = 5$ only 4.6% of the weight-four patterns do, and at $d = 7$ only 0.04% of the weight-six ones (Table SVII); lower bounds on optimal fault-tolerant cat preparations [31] therefore do not apply to real-T checks.

The double checks of Ref. [1] measure the transversal check with a GHZ fold over data and partner qubits, and a single X fault on a fold node spreads, after unfolding, to the data of a whole subtree: a hook. For the circuits of Ref. [1], $h = 1$ at $d = 3$ and $h = 2$ at $d = 5$, and every malignant configuration of minimal weight is of the form of Lemma 12, with the predicted acceptance $2^{-(\delta-1)}$ and bad weight (Table I). The bound of Corollary 13 is therefore attained.

Ref. [1] gives circuits up to $d = 5$, and how much fault distance is lost at larger distance was open [15]. We built GHZ-fold double checks of the same structure on the triangular color codes (Sec. S8): partners prepared in $|+\rangle$ and paired with the data, a CNOT tree that folds the check onto a root, which is measured and reset, the reverse unfold and unpairing, and a measurement of the partners, between flagged stabilizer rounds.

Proposition 14 (computer-assisted)**.** *On the* $[[7, 1, 3]]$*,* $[[19, 1, 5]]$ *and* $[[37, 1, 7]]$ *color codes, these GHZ-fold double checks have real-T fault distance two, three and four, for each fold tree we tested (three trees at* $d = 3$ *and* $d = 5$*, six at* $d = 7$*, of depth one to seven), while their proxy fault distance is three, five and at least five.*

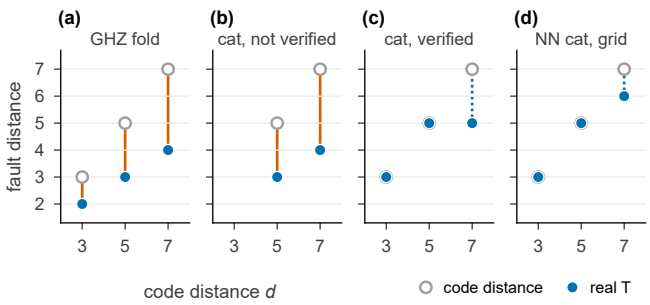


FIG. 3. Real-T fault distance of cultivation checks on the triangular color codes, certified exactly (filled), against the code distance (open), which bounds every fault distance (Proposition 21) and is reached by the S proxy at $d = 3$ and 5. The gap (vermillion) is the distance lost to real T gates. (a) GHZ folds, Proposition 14 (at $d = 3$ and 5 also the circuits of Ref. [1]); (b), (c) cats without and with verification, Proposition 16; (d) nearest-neighbour cats on a square grid, Proposition 17. At $d = 7$ the verified cats are certified to at least five and six, and the dotted range is open.

At $d = 3$ and 5 the construction reproduces the fault distances of the circuits of Ref. [1]. At $d = 7$, every harmful error found at weight four is realized by four fault classes whose exact detector patterns cancel. The real-T fault distance of the fold grows with the code distance, at half its rate (Fig. 3). Its failures are, however, strongly suppressed: by Lemma 12 a coherent completion of X on $|u|$ data qubits is accepted with probability $2^{-|u|}$, and at $d = 7$ every malignant configuration of four faults is accepted with probability between 2^{-10} and 2^{-4}. The leading term of the probability of an accepted wrong output ranges from $0.011\,p^4$ to $15\,p^4$ over the six trees, so the choice of tree, which costs nothing, changes it by three orders of magnitude, while the loss of distance remains.

The lemma also limits repairs. The completing fault can occur anywhere after the unfold, including in any layer inserted there, so flags must catch the hooks themselves. A hook on a fold qubit that is a CNOT target is detected only by a two-qubit parity $Z_a Z_b$, which no single-qubit flag measures without disturbing the check, and such placements exist in the $d = 5$ double check:

Proposition 15 (computer-assisted)**.** *No set of non-disturbing single-qubit Z- or X-type flags whose two couplings lie in the double check before the unpairing layer raises the real-T fault distance of the* $d = 5$ *double check of Ref. [1] above three. Two two-qubit flags raise it to four.*

With the two flags, the logical error falls by 2.2 and 3.5 times at $p = 10^{-3}$ and 5×10^{-4} in the SI1000 model [32], at a cost of 2–5% in acceptance (Sec. S8). The hand-designed flags of Ref. [15] also reach four in the full protocol, with 13 two-qubit flags, and five with a fourth stabilizer round [15, 16]. Without that round, fault distance

five requires flagging data errors that enter the check before its first T layer, and the flag sets our integer programs find for it grow to tens of flags and layers.

A CHECK THAT KEEPS THE FAULT DISTANCE

Corollary 13 suggests the remedy: let no single fault place X on more than one data qubit before the last T layer. A Shor-style check [25] does this. Each check round applies the $U^\dagger$ layer, prepares a GHZ state on fresh qubits c_q, verifies it by measuring all adjacent parities $Z_{c_q}Z_{c_{q+1}}$ before it touches the data, couples it once to each data qubit by $CX_{c_q,q}$, measures the cat in the X basis, whose parity is the check outcome, and applies the U layer. A fault in the cat preparation that spreads X over several cat qubits is caught by the verification, and every other fault reaches at most one data qubit. Flagged stabilizer rounds [29] surround each check round.

Proposition 16 (computer-assisted)**.** *On the* $[[19, 1, 5]]$ *color code, the verified-cat check has fault distance exactly five with real T gates, with one check round and with two, and with one round the probability of an accepted wrong output is* $3452\,p^5 + O(p^6)$ *under uniform circuit noise. On the* $[[37, 1, 7]]$ *code, one round has fault distance at least five. Without verification, the same circuits have fault distance three at* $d = 5$ *and four at* $d = 7$*. On the* $[[7, 1, 3]]$ *code, the verified-cat check keeps fault distance three, where the double check of Ref. [1] has two.*

The certificates exclude every malignant configuration of at most four faults, and the upper bounds come from Proposition 21; the exact enumeration of all 754,515 malignant configurations of five faults gives the leading coefficient. The weight-four certificates evaluate 3.2 million distinct errors E for two rounds at $d = 5$ and half a million at $d = 7$. The S proxy does not distinguish the verified and unverified circuits; with real T gates their fault distances differ by two at $d = 5$ (Table I). The unverified checks we certified, the GHZ folds at $d = 3$, 5 and 7 and the unverified cat at $d = 5$ and 7, all have fault distance $(d+1)/2$. The price of verification is acceptance, about half that of the unverified cat under uniform noise, and connectivity: the cat check as written is not nearest-neighbour on a square grid (the nearest-neighbour checks below remove both costs). Acceptance also limits the distance itself. In our circuits, under uniform noise at $p = 10^{-3}$, going from $d = 5$ to $d = 7$ lowers the acceptance from 5% to 0.05% for the GHZ fold, from 23% to 1.5% for the unverified cat and from 11% to 0.09% for the verified cat; at $p = 5 \times 10^{-4}$ the values at $d = 7$ are 2.0%, 12% and 2.9% (Table SVI). Our circuits are not optimized for acceptance; the published $d = 5$ protocol accepts 14% at $p = 10^{-3}$. At $d = 3$ fault distance three is also expensive in its coefficient: the verified cat has 2,405 malignant triples and a leading term $4.2 \times 10^3 p^3$, larger than the $0.43p^2$ of the double check of Ref. [1] for $p \gtrsim 10^{-4}$.

TABLE I. Real-T fault distance of cultivation checks, certified exactly with Theorems 2 and 5, and the leading term of the probability of an accepted wrong output under uniform circuit noise of strength p. The proxy distance is that of the same circuit with S in place of T. Malignant: number of malignant configurations of minimal weight; those of the first two rows are accepted with probability $2^{-(\delta-1)}$ relative to the fault-free run. The first three rows are the protocol of Ref. [1]; the $d = 7$ GHZ-fold row is our construction of Proposition 14, and the cat rows are one check round (two where stated); both sit between flagged stabilizer rounds on an ideally encoded input. For the $d = 7$ GHZ fold, the ranges are over the six trees. The nearest-neighbour (NN) checks of Proposition 17 have one check round, an un-grown cat and no stabilizer rounds; the NN protocol prepares the state itself with every operation noisy; the last row is the two-check circuit of Ref. [2] with its injection and error-correction cycles, under SI1000 noise (Proposition 18). Fault distances are at most the code distance (Proposition 21).

check	d	proxy	real T	malignant	leading term
GHZ fold [1]	3	3	2	9	$0.43\,p^2$
GHZ fold [1]	5	5	3	9	$1.5\,p^3$
GHZ fold, 2 flags	5	5	4	6824	$1.2 \times 10^3 p^4$
GHZ fold, 6 trees	7	≥ 5	**4**	15–957	$(0.011\text{–}15)\,p^4$
cat, not verified	5	5	3	99	$0.54\,p^3$
cat, verified	5	5	**5**	754,515	$3.5 \times 10^3 p^5$
cat, verified, 2 rounds	5	5	**5**		
cat, not verified	7	≥ 5	4	120	$0.093\,p^4$
cat, verified	7	≥ 5	**5–7**		
cat, verified	3	3	3	2405	$4.2 \times 10^3 p^3$
NN cat, ring	3	3	**3**	574	$3.6\,p^3$
NN cat, ladder	5	5	**5**	85,464	$12\,p^5$
NN cat, ladder	7	≥ 5	**6–7**		
NN protocol	3	3	**3**	10,573	$2.8 \times 10^3 p^3$
Google experiment	3	3	2	20	$2.2\,p^2$

NEAREST-NEIGHBOUR CHECKS AND THE EXPERIMENT

On a square grid, a cat state can be verified with the same economy if it is grown along a path of grid sites from the middle outward. A single X fault on a path qubit then spreads to the part of the path beyond it, which contains one of the two ends, or stays on that qubit, so the parity Z_aZ_b of the two ends, measured through an ancilla next to both, detects every spread. Each data qubit sits next to its own path site and interacts with the check once (Fig. 4). After the coupling the cat is un-grown onto its root by the same CNOTs in reverse: the check outcome is then a single X measurement instead of the parity of the whole path, and the other path qubits return to $|0\rangle$,

where their measurements are further detectors; faults in this step touch no data qubit. At $d = 3$ the path runs around a rectangle and the ancilla closes it. At $d = 5$ it is folded into a U on three rows, and the middle row measures rungs $Z_{(x,0)}Z_{(x,2)}$, each through one ancilla, and diagonals $Z_{(x,0)}Z_{(x-1,2)}$, each through two adjacent ancillas. Rungs detect any spread confined to one arm of the U; diagonals detect the equal spreads on both arms that two faults at mirror positions produce, and without them the ladder has fault distance four. Neither check needs stabilizer rounds: the code is checked only by the final ideal decoder, as in the circuits of Ref. [1].

Proposition 17 (computer-assisted)**.** *With two-qubit gates only between neighbours of a square grid, the ring check on the* $[[7, 1, 3]]$ *code (17 qubits) and the ladder check on the* $[[19, 1, 5]]$ *code (46 qubits) have real-T fault distance three and five, equal to the code distance. The probability of an accepted wrong output is* $3.6\,p^3$ *and* $12\,p^5$ *under uniform noise, at* $d = 3$ *the same as with the S proxy; under SI1000 noise it is* $0.056\,p^3 + 30\,p^4$ *at* $d = 3$*, and its leading term is* $0.012\,p^5$ *at* $d = 5$*. At* $p = 10^{-3}$ *in SI1000 the checks accept 88% and 39% of runs. The ladder on the* $[[37, 1, 7]]$ *code (91 qubits) has fault distance at least six.*

Without un-growing, the X measurements of the cat dominate the failures, since each one flips the check outcome and fails with probability $5p$ in SI1000: the leading terms are then $6.9\,p^3$ and $43\,p^5$, at acceptance 89% and 55%. These are the cheapest checks we found, by every measure in Table II. At $d = 5$ the ladder uses a third of the qubits of the verified cat of Proposition 16 and at most two thirds of its depth, accepts 13 to 18 times more often, and its leading term is 300 times smaller under uniform noise; most of the difference is the stabilizer rounds, which the ladder does not need. At $d = 7$ the ladder accepts 7 to 21% of runs, where our checks with stabilizer rounds accept 3×10^{-6} to 10^{-3}. At $d = 3$ the ring is ahead of every GHZ fold at every $p < 10^{-2}$, with half the qubits and twice the acceptance of the best one. Fault distance five at $d = 5$ costs less than three: the ladder accepts four to six times more often than the unverified cat and 20 to 28 times more often than the best fold, at a logical error three orders of magnitude lower. In SI1000 the minimal failures of the un-grown checks consist of faults of strength $p/10$, so the next order matters at $p = 10^{-3}$; at $d = 3$ it adds the $30\,p^4$ above.

Google's experiment. The cultivation experiment of Ref. [2] ran a $d = 3$ color code on a square grid with nearest-neighbour CZ gates, injection, GHZ-fold checks with time-reversed unfolding, and error-correction cycles between checks, and projected its logical error with the S proxy. We converted the published circuits [2] into real-T circuits, with each physical $R_y(\pm\pi/4)$ rotation written as T or $\mathrm{T}^\dagger$ conjugated by the circuit's own H_{YZ} gates and the injection sign chosen so that the fault-free check accepts with certainty, and added SI1000 noise, the model of the authors' simulations, with noiseless virtual T gates.

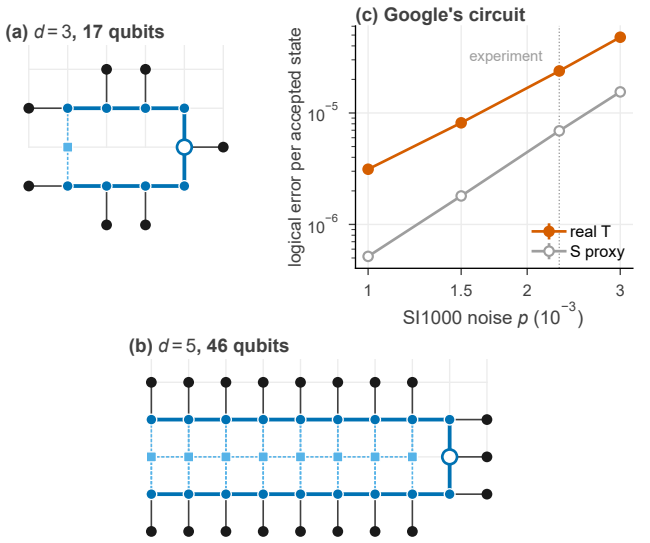


FIG. 4. (a), (b) Nearest-neighbour verified-cat checks of Proposition 17 on a square grid: data qubits (black), the cat path (blue) grown from its middle (open circle), and verification ancillas (light blue) with the parities they measure (dashed). (c) Logical error per accepted state of Google's cultivation circuit [2] (two checks with error-correction cycles) under SI1000 noise, with real T gates (vermillion) and with the S proxy (grey), at equal acceptance; the experiment ran at $p \approx 2.3 \times 10^{-3}$ (dotted).

Proposition 18 (computer-assisted)**.** *After one, two or three checks, the circuit of Ref. [2] has real-T fault distance two, while its proxy fault distance is three. After two checks, each of its 20 malignant pairs contains a fault in the unfolding of the last check, and an ideal injection leaves the same 20 pairs.*

The experiment is thus a GHZ fold with fault distance $(d+1)/2$, as Proposition 14 predicts. At the noise strength of the experiment, $p = 2.3 \times 10^{-3}$, the two-check circuit accepts 22% of runs and its logical error is 2.4×10^{-5} with real T gates against 6.9×10^{-6} with the proxy; the ratio grows from 3.1 at $p = 3 \times 10^{-3}$ to 4.5 at 1.5×10^{-3} and 6.1 at 10^{-3}, because the leading term is $2.2\,p^2$ instead of order p^3 (Fig. 4). Projections of the experiment to lower noise along the p^3 slope of the proxy are therefore optimistic, by a factor that grows as $1/p$. On the same kind of grid and from the same ideal input, the two checks of the experiment have leading term $0.44\,p^2$, and one ring check of Proposition 17 has $0.056\,p^3 + 30\,p^4$, about 10^{-3} times the failure probability at $p = 2.3 \times 10^{-3}$, with 17 qubits instead of 14.

A complete protocol. The checks above start from an ideally encoded state. On the grid the state can also be prepared with every operation noisy. A $\mathrm{T}^\dagger$ gate is applied

TABLE II. Cost of cultivation checks under SI1000 noise at $p = 10^{-3}$: qubits, circuit layers, acceptance relative to the fault-free run, real-T fault distance (FD), leading terms of the probability of an accepted wrong output, the resulting logical error per accepted state ϵ, and the expected spacetime volume per accepted state, qubits × layers / acceptance. Rows marked * prepare the state themselves (injection, and growth at $d = 5$, for Refs. [1, 2], whose ϵ is sampled; relayed injection for the NN protocol); the other rows start from an ideally encoded state. NN: nearest-neighbour on a square grid; GHZ fold: the tree with the smallest leading term. At fault distance five, ϵ is the leading term only, and it is omitted where the next order dominates.

check	d	qubits	layers	acc.	FD	leading terms	ϵ	volume
Ref. [1]*	3	15	39	0.72	2		8.0×10^{-7}	8.1×10^{2}
Google*, NN	3	14	156	0.52	2	$2.2\,p^2$	3.1×10^{-6}	4.2×10^{3}
GHZ fold	3	38	46	0.43	2	$0.049\,p^2$	4.9×10^{-8}	4.1×10^{3}
cat, verified	3	69	34	0.22	3	$1.7 \times 10^{4} p^3$	1.7×10^{-5}	1.1×10^{4}
NN ring, not un-grown	3	17	14	0.89	**3**	$6.9\,p^3$	6.9×10^{-9}	$\mathbf{2.7 \times 10^{2}}$
NN ring	3	17	15	0.88	**3**	$0.056\,p^3 + 30\,p^4$	$\mathbf{8.6 \times 10^{-11}}$	2.9×10^{2}
NN protocol*	3	17	107	0.11	**3**	$7.8 \times 10^{3} p^3$	7.8×10^{-6}	1.7×10^{4}
Ref. [1]*	5	42	94	0.16	3		2.2×10^{-9}	2.5×10^{4}
GHZ fold	5	110	74	0.020	3	$0.052\,p^3$	5.2×10^{-11}	4.1×10^{5}
cat, not verified	5	110	36	0.093	3	$0.46\,p^3$	4.6×10^{-10}	4.2×10^{4}
cat, verified	5	128	51	0.030	5	$8.3 \times 10^{3} p^5$	8.3×10^{-12}	2.1×10^{5}
NN ladder, not un-grown	5	46	24	0.55	**5**	$43\,p^5$	4.3×10^{-14}	$\mathbf{2.0 \times 10^{3}}$
NN ladder	5	46	34	0.39	**5**	$0.012\,p^5$		4.0×10^{3}
GHZ fold	7	218	114	3×10^{-6}	4	$0.0087\,p^4$	8.7×10^{-15}	8×10^{9}
cat, not verified	7	218	51	9.1×10^{-4}	4	$0.086\,p^4$	8.6×10^{-14}	1.2×10^{7}
cat, verified	7	254	88	1×10^{-5}	≥ 5			2×10^{9}
NN ladder, not un-grown	7	91	33	0.21	≥ 5			$\mathbf{1.5 \times 10^{4}}$
NN ladder	7	91	52	0.068	≥ 6			7.0×10^{4}

to one data qubit in $|+\rangle$, and the CNOTs of a non-fault-tolerant encoder are relayed along the path: the control is copied onto its path site, spread along the segment that covers its targets, applied, and unspread. The encoder's hooks are single faults that leave X or Z on several data qubits, so all six stabilizers are then measured, each with its own verified path cat, before three checks. This 17-qubit protocol has real-T fault distance three, equal to its proxy distance; without the stabilizer measurements, or with only the Z stabilizers, it has two, because a Z hook of weight two is completed by one more fault into a logical Z that no check sees. Its cost is that of the stabilizer measurements: 107 layers with many measurement layers, during which every data qubit idles at $2p$ in SI1000, so its leading term is $7.8 \times 10^{3} p^3$ at 11% acceptance, ahead of the experiment's circuit for $p \lesssim 3 \times 10^{-4}$ (Table II). Preparing $|0\rangle_L$ by measuring the X stabilizers and letting the first check project onto the T state, which needs no T gate, also reaches fault distance three, at a higher cost. At $d = 3$ on a grid, fault distance is thus no longer the obstacle; the cost of measuring stabilizers around the checks is.

OUTLOOK

Theorem 2 reduces the fault tolerance of a cultivation circuit to a stabilizer code on its T gates and a binary matrix, the columns of D and E, and Theorem 8 splits the fault distance into the order of the code, a question of distillation, and the spreading of faults by the measurement circuits. Measured in T gates, cultivation costs exactly as much as distillation, so every lower bound on one is a lower bound on the other. Because the checks of cultivation are deterministic, Theorem 5 decides the collapse condition by $\mathbb{Z}_4$ Gauss sums, which makes exact fault distances and leading terms available where sampling with real T gates [9–13, 33, 34] cannot reach. With them, the spreading of faults turns out to be the limiting factor in practice. Real T gates turn undetected X errors on part of a logical operator before the last T layer into coherent errors that further faults complete into a logical error (Lemma 12). GHZ-fold and unverified-cat checks spread such errors from single faults and keep fault distance $(d + 1)/2$ in every case we certified, up to $d = 7$, and so does the circuit run on hardware. A check that is fault tolerant relative to the code keeps the full distance, and on a square grid a cat grown from the middle of a path and verified at its ends, rungs and diagonals does so with fewer qubits and higher acceptance than the GHZ

folds of current protocols.

Our model assumes stochastic Pauli noise on every operation, including the T gates; for coherent errors, whose fault configurations interfere, the criterion applies to each term but not to their sum, and leakage, which the experiment also sees [2], is outside it. Proposition 1 gives up the acceptance probability, so the lower bounds concern the size of a single run rather than the expected cost per accepted state. The checks of Propositions 14, 16 and 17 start from an ideally encoded state; the complete nearest-neighbour protocol shows that the preparation can keep the full distance, and its cost shows that cheap stabilizer measurement on the grid, not the T checks, is what a competitive protocol still needs. Open questions include lower bounds on qubits, two-qubit gates or time steps at fault distance two; the least number of T gates at fault distance three, between ten and fifteen; lower bounds on the gates of real-T checks, which the code-relative condition of Corollary 13 and the linear-algebraic form of Theorem 5 turn into a constraint problem, as is done for cat states [31]; and whether nearest-neighbour checks keep the full distance beyond $d = 5$, where the ladder is certified to at least six.

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Data availability. The scripts, circuits and results are available at `https://github.com/llorebaga/Bounds_magic`; the exact evaluation of Theorem 5 and the circuits of Propositions 14–18 are in its directory `cultivation_exact`. The circuit chunks and state-vector data of Ref. [28] are reproduced there under the Apache License 2.0, and the circuits of Ref. [2] are decoded from the links published with that paper.

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Supplemental Material

S1. PROOFS

Proof of Proposition 1. For a fixed fault configuration, the unnormalized accepted output of the protocol is $\sum_{b'} q\, K_{b',F'} \rho_0 K_{b',F'}^\dagger$, summed over accepted runs b' and the faults F' along them, with probabilities $q > 0$; this is exact for stochastic Pauli noise, because faults are sampled classically at the locations actually executed. A run (b, F) is harmless exactly when its own term is a multiple of $|\tau\rangle\langle\tau|$. Fix an accepted fault-free run b. Along b, every random choice and every measurement outcome is fixed, so the operations executed along b form a non-adaptive circuit, and for every fault configuration F at its locations, $K_{b,F}$ is the Kraus operator of this circuit with F inserted, postselected on the outcomes of b. A discarded qubit contributes a partial trace, $\sum_k \langle k| \cdot |k\rangle$; if the sum is a multiple of a pure state, each term is, so replacing the partial trace by the removal $\langle k|$ for an outcome k with $\langle k|K_{b,\emptyset}|0\rangle \neq 0$ keeps the circuit correct and every harmless configuration harmless. Hence every configuration with $\mathrm{wt}(F) < d$ is harmless in the circuit. The removals are ideal and add no locations. □

Proof of Theorem 2. Skeleton. Eq. (2) follows from $\mathrm{CX}_{q\to a}(\alpha|0\rangle + \beta|1\rangle)(|0\rangle + \omega|1\rangle)/\sqrt{2}$, whose component with a in $|0\rangle$ is $(\alpha|0\rangle + \beta\omega|1\rangle)/\sqrt{2}$. Introduce one ancilla a_j per T gate, prepared ideally in $|T\rangle$, and write $\mathrm{T}^\dagger = S^\dagger T$ with an ideal $S^\dagger$. A fault of the jth T gate acts after the gadget. The circuit becomes $K_F = 2^{t/2} K_{S,F}(|0\rangle \otimes |T\rangle^{\otimes t})$, where the skeleton $K_{S,F}$ consists of Clifford gates, projections and removals, and F is inserted at the same qubits and times. Let $M_F = K_{S,F}(|0\rangle \otimes \cdot)$, a linear map from t qubits to r qubits, and $M = M_0$. As in the proof of Theorem 1 of Ref. [17], M applied to half of t Bell pairs gives a stabilizer state on $t + r$ qubits, and by the local-Clifford normal form of bipartite stabilizer states $M = c\, V \Pi_C$ with a stabilizer code C on t qubits and a Clifford isometry V from C onto its range. The range contains $|\tau\rangle$ and is a stabilizer subspace; since no Pauli operator other than the identity fixes τ, it is the whole output space, so $\dim C = 2^r$ and V is unitary from C onto $(\mathbb{C}^2)^{\otimes r}$.

Signs. Treat each removal $\langle k|_q$ as a projection of Z_q with sign $(-1)^k$ followed by an ideal removal, and let $\sigma \in \mathbb{F}_2^\mu$ list the signs of all μ projections relative to the circuit, so that $M^{(0)} = M$. Let $|\Psi^{(\sigma)}\rangle = (M^{(\sigma)} \otimes I) \sum_x |x\rangle|x\rangle$ be the unnormalized Choi vector. Following the circuit with the stabilizer formalism [19], the Choi vector stays a pure stabilizer state, so the Pauli of every projection either anticommutes with a current stabilizer or is one up to sign. In the first case the projection halves the squared norm and replaces that stabilizer by the projected Pauli with the chosen sign, and in the second it keeps the state or annihilates it, depending on the sign. The unsigned stabilizer groups and the norms of the nonzero $|\Psi^{(\sigma)}\rangle$ are therefore independent of σ. The signs required by the deterministic projections are affine functions of the earlier signs, so $|\Psi^{(\sigma)}\rangle \neq 0$ exactly when $D_0\sigma = 0$ for a linear map D_0. Two stabilizer states with the same unsigned group and norm differ by a Pauli operator and a phase, and the Pauli is determined by the sign difference modulo the stabilizer group, linearly. Hence, for $D_0\sigma = 0$, $|\Psi^{(\sigma)}\rangle = \zeta_\sigma(A_\sigma \otimes B_\sigma)|\Psi^{(0)}\rangle$ with $|\zeta_\sigma| = 1$ and Paulis A_σ on the output and B_σ on the reference, that is, $M^{(\sigma)} = \zeta_\sigma A_\sigma M B_\sigma^{\mathsf{T}}$.

Faults. A Pauli P inserted after an operation is moved to the end of the circuit: a Clifford gate G turns it into $GPG^\dagger$, and a projection $(I + sQ)$ with which it anticommutes becomes $(I - sQ)$ as P passes it. A removal absorbs the Z part of P on the removed qubit, and its projection turns the X part into a sign flip. A sign flip changes σ directly. Hence $M_F = \zeta\, P_{\mathrm{out}}(F)\, M^{(s(F))}$ with a phase ζ, a Pauli $P_{\mathrm{out}}(F)$ on the output and a sign vector $s(F)$, both linear in F. Set $D(F) = D_0 s(F)$, where D_0 has one row per projection, nonzero only for the projections whose outcome is fixed in the skeleton. If $D(F) \neq 0$, then $M_F = 0$ and the configuration is rejected. Otherwise $M_F = \zeta' P_{\mathrm{out}} A_s M B_s^{\mathsf{T}}$. For an output Pauli R, write $V\Pi_C = W(I \otimes \langle 0|^{\otimes(t-r)})U\Pi_C$ with Clifford unitaries U on t qubits, mapping C onto $(\mathbb{C}^2)^{\otimes r} \otimes |0\rangle^{\otimes(t-r)}$, and W on r qubits; then $RM = ML$ with the Pauli $L = U^\dagger(W^\dagger R W \otimes I)U$, which commutes with the stabilizers of C. Hence $M_F = \zeta'' M\, E(F)$ with $E(F) = L(P_{\mathrm{out}} A_s) B_s^{\mathsf{T}}$. Finally, $ME \propto ME'$ holds exactly when $E'E^\dagger$ acts on C as a scalar, that is, lies in $\langle iI, S_C\rangle$, because V is injective on C and a Pauli that anticommutes with a stabilizer maps C to an orthogonal code. So E is well defined and linear modulo $\langle iI, S_C\rangle$.

Collapse. $K_F|0\rangle = 2^{t/2}\zeta'' c V \Pi_C E(F)|T\rangle^{\otimes t}$ and $K_0|0\rangle \propto V\Pi_C|T\rangle^{\otimes t} \propto |\tau\rangle$. Since V is injective on C, $K_F|0\rangle \in \mathbb{C}|\tau\rangle$ exactly when Eq. (1) holds.

Z faults at T gates. A Z fault after the gadget of the jth T gate on qubit q satisfies $Z_q\langle 0|_a \mathrm{CX}_{q\to a} = \langle 0|_a Z_a Z_q \mathrm{CX}_{q\to a} = \langle 0|_a \mathrm{CX}_{q\to a} Z_a$, using $\langle 0|_a = \langle 0|_a Z_a$ and $\mathrm{CX}_{q\to a} Z_a = Z_q Z_a \mathrm{CX}_{q\to a}$. It acts as Z on the ideal input a_j, so $D = 0$ and $E = Z_j$. A Z fault before the gate is the same, because Z commutes with T and with $S^\dagger$. □

Proof of Lemma 3. (i) On one qubit, $X|T\rangle = (|T\rangle + iZ|T\rangle)/\sqrt{2}$, $XZ|T\rangle = -(i|T\rangle + Z|T\rangle)/\sqrt{2}$ and $Y = iXZ$, so up to a phase $X^u Z^v |T\rangle^{\otimes t} = \sum_w \gamma_w Z^{v+w}|T\rangle^{\otimes t}$, summed over w with $\mathrm{supp}\, w \subseteq \mathrm{supp}\, u$ and with $|\gamma_w| = 2^{-|u|/2}$. If each projected term lies on the line $\mathbb{C}\,\Pi_C|T\rangle^{\otimes t}$, so does their sum, and Theorem 2 applies. (ii) Multiplying E by an

element of $\langle iI, S_C\rangle$ changes $\Pi_C E|T\rangle^{\otimes t}$ only by a phase, so we may take $E = X^u Z^v$ acting on fewer than d qubits. Every pattern $v+w$ in (i) is supported on these qubits and so has weight below d, and (i) applies. Now let E commute with S_C and $|E|_S < d$. Then E commutes with Π_C, and $E\Pi_C|T\rangle^{\otimes t} = \Pi_C E|T\rangle^{\otimes t}$ is a multiple of $\Pi_C|T\rangle^{\otimes t}$, nonzero because E is unitary. Through the isometry V, E acts on C as a Pauli operator P on r qubits up to a phase, and $|\tau\rangle$ is an eigenvector of P. A Hermitian multiple P' of P has eigenvalues ± 1, so P' or $-P'$ fixes $|\tau\rangle$; since $\nu(\tau) = r$, $P' = \pm I$. Hence E acts on C as a scalar, that is, $E \in \langle iI, S_C\rangle$, and every logical operator outside $\langle iI, S_C\rangle$ acts on at least d qubits. □

Lemma 19 (Checked cat-state measurement)**.** *Let a Pauli operator g act on w data qubits with letters $P_1, \dots, P_w$. Prepare $(|0\rangle^{\otimes w} + |1\rangle^{\otimes w})/\sqrt{2}$ on cat qubits $c_1, \dots, c_w$ by preparing c_1 in $|+\rangle$ and the others in $|0\rangle$ and applying CNOTs from c_1; in each of ℓ rounds, measure $Z_{c_1} Z_{c_i}$ for $i = 2, \dots, w$ with a fresh ancilla in $|0\rangle$, the target of CNOTs from c_1 and c_i, and keep the outcome $+1$; apply controlled-P_i from c_i to the ith data qubit for every i; measure every c_i in the X basis and keep the runs whose outcomes have product s. If at most ℓ faults occur in this gadget and the run is kept, it acts on the data as $R\,(I + s'g)/2$ up to a nonzero scalar, with $s' = \pm s$ and a Pauli operator R for which R or Rg acts on at most as many qubits as there are faults.*

Proof. Without faults, X outcomes with product $(-1)^{|x|}$ leave a nonzero multiple of $[I + (-1)^{|x|} g]/2$ on the data. Push every fault to the end of the gadget. Z errors on cat qubits, which Z errors on the ancillas also produce, and flips of the X measurements change the product of the X outcomes; X errors on the ancillas and flips of their measurements change only the checks. A Pauli on a data qubit before its controlled Pauli can be moved after it, at the cost of a Z error on its cat qubit when it anticommutes with P_i. Let $x \in \mathbb{F}_2^w$ mark the X errors on the cat qubits at the controlled Paulis; they reach the data as $P^x = \bigotimes_i P_i^{x_i}$, and $P^{x+\mathbf{1}} = \pm g P^x$. During the rounds the cat qubits are only controls of CNOTs, so a fault there changes at most one entry of x, as does a fault on a cat qubit before its own controlled Pauli, and a fault in a controlled Pauli or on a data qubit adds a Pauli on one data qubit, since an X error on c_i after its controlled Pauli no longer spreads. If the preparation of the cat state is fault-free, x starts at $\mathbf{0}$. Otherwise at most $\ell - 1$ faults fall into the ℓ rounds, so one round is fault-free; it reads $x_1 + x_i$ correctly for every i, so a kept run has $x \in \{\mathbf{0}, \mathbf{1}\}$ after it, and only later faults change x. In both cases the data error is R or $\pm Rg$ with R acting on at most as many qubits as there are faults. □

Proof of Theorem 8. Bounds. Choose a pattern e of weight d_C that violates the distillation condition. Z faults at the corresponding T gates have $D = 0$ and $E = Z^e$ (Theorem 2), so they are harmful, and $d \le d_C$. If $\mathrm{wt}(F) < \min(d_C, d_{\mathrm{sp}})$ and $D(F) = 0$, then $|E(F)|_S \le \mathrm{wt}(F) < d_C$, and F is harmless by Lemma 3(ii).

Construction. Let $g_1, \dots, g_m$ generate S_C, with signs s_k. The circuit prepares t data qubits in $|+\rangle$, applies T to each, measures $g_1, \dots, g_m$ in turn with the gadget of Lemma 19, with $\ell = d_C - 1$ and $s = s_k$, and passes the data to the ideal decoder of C, which projects onto $(I + s_k g_k)/2$ for every k and unencodes. Without faults the checks of the cat states are deterministic and the measurements apply Π_C, so the output is a nonzero multiple of $\Pi_C|T\rangle^{\otimes t}$, the circuit is correct, and the decoder is admissible. In the skeleton, the gadget of a T gate on $|+\rangle$ maps its input to the data qubit identically, $\sqrt{2}\langle 0|_a \mathrm{CX}_{q\to a}|+\rangle_q|\psi\rangle_a = |\psi\rangle_q$, so the code of the circuit is C. Let F have fewer than d_C faults, and let every check of the cat states pass. By Lemma 19 the kth measurement acts as $R_k(I + s'_k g_k)/2$, with R_k or $R_k g_k$ acting on at most as many qubits as there are faults in that gadget, and every other fault, on a T gate or an idle data qubit, is a Pauli operator on one data qubit; so is a fault on the preparation of $|+\rangle$, which leaves $|+\rangle$ or $|-\rangle$, with $T|-\rangle = ZT|+\rangle$. Move all these Paulis to the T inputs. A Pauli R moved past $(I + s'g)/2$ turns it into $(I + (-1)^{\langle R,g\rangle} s' g)/2$, so the circuit applies $\prod_k (I + s_k g_k)/2 \cdot \prod_k (I + s''_k g_k)/2 \cdot R\,|T\rangle^{\otimes t}$ with signs s''_k, where R is the product of all the Paulis. Since the projections commute, this vanishes unless $s''_k = s_k$ for every k, and it is then a multiple of $\Pi_C R|T\rangle^{\otimes t}$. Replacing R_k by $R_k g_k$ changes R only within $R\langle iI, S_C\rangle$, so $|R|_S \le \mathrm{wt}(F) < d_C$, and Lemma 3(ii) shows that F is harmless. Faults that fail a check are rejected. Hence the fault distance is at least d_C, and by the bound above it equals d_C. □

Proof of Corollary 9. By Corollary 7, every accepted run executes at least $n_{\min}(\tau, d)$ T gates. By Theorem 1 of Ref. [17], an order-d distillation protocol on $n = n_{\min}(\tau, d)$ inputs yields a stabilizer code C on n qubits whose projection of $|T\rangle^{\otimes n}$ encodes τ and satisfies the distillation condition for all patterns of weight below d, so $d_C \ge d$. Theorem 8 turns C into a circuit with n T gates and fault distance $d_C \ge d$. By Lemma 3(ii), C, and the code of every order-d distillation protocol, has r logical qubits and distance at least d. Such a code corrects $\lfloor (d-1)/2 \rfloor$ errors, and by Ref. [26] its length is at least $6\lfloor (d-1)/2 \rfloor - 1$, which is $3d - 4$ for odd d. For even d, feeding one input of an order-d protocol with $I/2$ gives an order-$(d-1)$ protocol [17], so $n_{\min}(\tau, d) \ge n_{\min}(\tau, d-1) + 1 \ge 3d - 6$. □

Proof of Proposition 10. Let $g_1, \dots, g_8$ be the measured stabilizers in their order (Table SIII); they generate the stabilizer group S of the factory code, which is the code C of Theorem 2 for this circuit, with the T gates numbered

by their qubits. A fault after a preparation or a T gate is, up to stabilizers of $|+\rangle$, a Pauli of weight one on the T inputs, with no flips. Its Z part collapses, by the exact certificate of the single-fault minors in Ref. [17], and so does its X part, by Lemma 3(i). Consider a fault in the measurement of g_k, including its ancilla and flag. Pushed to the end of the measurement circuit, it becomes a Pauli R on the data, a flip ϕ of the outcome of g_k and a flip ψ of the flag. If $\psi = 1$, the flag detects it. If R anticommutes with some g_j, $j < k$, the decoder detects it, because g_j was projected before R and is projected again after it, with only commuting projections in between. If ϕ differs from the commutator $\langle R, g_k\rangle$, the decoder detects it, because the projection onto g_k then has the wrong sign relative to the decoder's. Otherwise R commutes with $g_1, \ldots, g_{k-1}$ and, together with the flip, equals R applied before the measurement of g_k; it can then be moved to the inputs, and the fault is harmless exactly when R collapses. An exact computation over $\mathbb{F}_2$ (`scripts/cultivation_t9.py`) sorts the 573 faults in the measurement circuits of Table SIII: 37 raise the flag, 288 leave an R that anticommutes with an earlier stabilizer, 152 flip the outcome inconsistently with R, and the remaining 96 leave an R of weight at most one modulo S, which collapses, since multiplication by a stabilizer changes $\Pi_C R|T\rangle^{\otimes 9}$ only by a sign. Measurement flips and faults on the decoder's input are covered by the same cases. □

Proof of Theorem 5. On one qubit, $|T\rangle = (|0\rangle + \omega|1\rangle)/\sqrt{2}$, $Z|T\rangle = (|0\rangle - \omega|1\rangle)/\sqrt{2}$ and $X|T\rangle = \omega(|0\rangle + \bar\omega|1\rangle)/\sqrt{2}$. Writing $|0\rangle + \bar\omega|1\rangle = \alpha(|0\rangle + \omega|1\rangle) + \beta(|0\rangle - \omega|1\rangle)$ gives $\alpha + \beta = 1$ and $\alpha - \beta = \bar\omega^2 = -i$, so $\alpha = \bar\omega/\sqrt{2}$, $\beta = i\bar\omega/\sqrt{2}$ and $X|T\rangle = (|T\rangle + iZ|T\rangle)/\sqrt{2}$. Taking the tensor product over $\operatorname{supp} a$ and using $X^a Z^b = (-1)^{|a\wedge b|} Z^b X^a$,

$$X^a Z^b |T\rangle^{\otimes t} = (-1)^{|a\wedge b|}\, 2^{-|a|/2} \sum_{w \le a} i^{|w|} Z^{b+w} |T\rangle^{\otimes t}. \tag{S1}$$

Apply Π_C. By Definition 4, the terms with $b + w \notin \Sigma \cup (\ell + \Sigma)$ vanish. If $e \in c\ell + \Sigma$, then $e + c\ell \in \Sigma$ and $Z^{e+c\ell} = \sigma(e + c\ell)g$ with $g \in S_C$, and $\Pi_C g = \Pi_C$; since Z^ℓ commutes with S_C, it commutes with Π_C, so $\Pi_C Z^e |T\rangle^{\otimes t} = \chi(e + c\ell) Z^{c\ell} \Pi_C |T\rangle^{\otimes t}$. This proves Eq. (3) with $\mu = (-1)^{|a\wedge b|}$. The vector $Z^\ell \Pi_C |T\rangle^{\otimes t}$ has the same norm as $\Pi_C |T\rangle^{\otimes t}$, and V maps the two to multiples, of equal modulus, of $Z|T\rangle$ and $|T\rangle$, which are orthogonal. The acceptance and the bad weight follow, and the bad weight is positive exactly when $g_1 \neq 0$. □

Proof of Proposition 6. The set $\{w \le a : w + b \in c\ell + \Sigma\}$ is empty or an affine subspace $w_0 + \langle k_1, \ldots, k_m\rangle$ of $\mathbb{F}_2^{|a|}$, found by Gaussian elimination in $O(t^3)$. For binary vectors $v_0, \ldots, v_m$, inclusion and exclusion give $|v_0 + \cdots + v_m| = \sum_{S \neq \emptyset} (-2)^{|S|-1} |\bigwedge_{s\in S} v_s|$, and the terms with $|S| \ge 3$ vanish modulo four. With $\chi(w) = (-1)^{x\cdot w}$ and $2(x \cdot w) \bmod 4$ linear in z, the summand of Eq. (4) is $i^{q(z)}$ with

$$q(z) = q_0 + \sum_l L_l z_l + 2 \sum_{l<m} B_{lm} z_l z_m, \quad L_l = |k_l| - 2|w_0 \wedge k_l| + 2\, x \cdot k_l, \quad B_{lm} = |k_l \wedge k_m| \bmod 2. \tag{S2}$$

To sum out z_v, let N be the set of its neighbours ($B_{vm} = 1$) and $y = \bigoplus_{m\in N} z_m$. The sum over z_v is $1 + i^{L_v}(-1)^y$. If L_v is odd, this is $(1 + i)\, i^{3y}$ or $(1 - i)\, i^y$; the parity y, read as an integer modulo four, is $\sum_m z_m - 2\sum_{m<m'} z_m z_{m'}$, so q stays quadratic. If L_v is even, it is $2[y = L_v/2]$: if N is empty, the sum vanishes or is multiplied by two; otherwise the constraint expresses one variable of N as a parity of the others, which is substituted into q in the same way. Each step costs $O(m^2)$ and multiplies the modulus by $\sqrt{2}$ or 2, or returns zero, so $|g_c|$ is zero or a power of $\sqrt{2}$, and the acceptance $2^{-|a|}(|g_0|^2 + |g_1|^2)$ and bad weight $2^{-|a|}|g_1|^2$ are dyadic. We checked the procedure against direct summation on 400 random forms with up to eight variables. □

Proof of Lemma 12. For $U_q = T^{\pm 1}$, $U_q X U_q^\dagger = \bar\omega^{\pm 1} S^{\pm 1} X$ and $S^{\pm 1} = \omega^{\pm 1}(I \mp iZ)/\sqrt{2}$. The error immediately after the layer is therefore, up to a phase, $\prod_{q\in u}(I \mp iZ_q) X^u / 2^{|u|/2} = 2^{-|u|/2} \sum_{w\subseteq u} c_w Z^w X^u$ with $c_w = \prod_{q\in w}(\mp i)$. The faults after the layer give $X^{\ell\setminus u} Z^w X^u = \pm Z^w X^\ell$. The ideal decoder projects onto the reference syndrome. X^ℓ is a logical operator, so it commutes with all stabilizers; Z^w with $w \subseteq u \subsetneq \ell$ has $|w| < \delta$, so it is not a nontrivial logical operator and commutes with all X stabilizers only if it is a Z stabilizer, which by assumption happens only for $w = \emptyset$. Every other component is mapped by the projection to an orthogonal syndrome space and removed. The accepted component is $2^{-|u|/2} X^\ell$ applied to the fault-free state, of relative probability $2^{-|u|}$, and it decodes to $X_L|\tau\rangle$. For $|\tau\rangle = |T\rangle$, $|\langle T|X|T\rangle|^2 = 1/2$, so the output is wrong with probability 1/2; with Y in place of the last X, the output is $X_L Z_L |\tau\rangle$ up to a phase, orthogonal to $|T\rangle$. In the proxy, $SXS^\dagger = Y$ up to a phase, so the error after the layer is the Pauli $Z^u X^\ell$ up to a phase; Z^u with $\emptyset \neq u \subsetneq \ell$ is neither a stabilizer nor a logical operator, so it has a nontrivial syndrome and the configuration is rejected. □

Two further consequences. The complement symmetry of distillation [17] and the code distance of an ideal decoder give the following two statements; neither is needed in the main text.

Corollary 20. *If all Pauli expectations of τ are rational, a fault configuration F of a correct non-adaptive circuit is harmless if and only if $F + Z_{\mathcal{T}}$ is, where $Z_{\mathcal{T}}$ places a Z fault after every T gate.*

Proof. The automorphism σ of $\mathbb{Q}(\sqrt{2}, i)$ that flips the sign of $\sqrt{2}$ and fixes i maps ω to $-\omega$, so, applied entrywise, it maps T and $\mathrm{T}^\dagger$ to ZT and $ZT^\dagger$. It maps every Clifford gate to itself up to a global phase, because up to such a phase the entries of a Clifford gate are $2^{-k/2}$ times Gaussian integers, and it fixes projections, Pauli faults and $|0\rangle$. Hence σ maps $K_F|0\rangle\langle 0|K_F^\dagger$ to $K_{F+Z_{\mathcal{T}}}|0\rangle\langle 0|K_{F+Z_{\mathcal{T}}}^\dagger$, as σ commutes with complex conjugation. It fixes $|\tau\rangle\langle\tau|$, whose entries lie in $\mathbb{Q}(i)$ because its Pauli coefficients are rational. Since σ^2 is the identity, one of the two operators is a multiple of $|\tau\rangle\langle\tau|$ exactly when the other is. □

Proposition 21. *If a correct circuit ends with an admissible ideal decoder for an $[[m, r, \delta]]$ code, and each of the m qubits has an idle location immediately before the decoder, then its fault distance d is at most δ, and $m \geq r + 2(d-1)$.*

Proof. A logical Pauli of weight δ placed at the idle locations commutes with the stabilizers, passes the decoder and applies a nontrivial Pauli to the decoded state, which differs from $|\tau\rangle$ because $\nu(\tau) = r$. The inequality is the quantum Singleton bound [35]. □

S2. CODES ON THE T GATES

A combinatorial form of the collapse condition. For a Hermitian Pauli operator P on t qubits, $\langle T|^{\otimes t} P |T\rangle^{\otimes t}$ is $\pm 2^{-\mathrm{wt}(P)/2}$ if P contains no Z, with the sign of P as a product of the letters I, X and Y, and zero otherwise, because $\langle T|X|T\rangle = \langle T|Y|T\rangle = 1/\sqrt{2}$ and $\langle T|Z|T\rangle = 0$.

Proposition 22. *For a Pauli operator E on the T inputs and a Hermitian Pauli operator L that commutes with S_C, let*

$$A_E(L) = \sum_{g \in S_C} (-1)^{\langle E, Lg\rangle} \langle T|^{\otimes t} Lg |T\rangle^{\otimes t}, \tag{S3}$$

the signed count of the elements of LS_C without Z, weighted by $2^{-\mathrm{wt}/2}$ and by their commutation with E. Then E satisfies Eq. (1) if and only if $A_E(L)A_0(I) = A_E(I)A_0(L)$ for every such L. It suffices to take one L from each of the 4^r cosets of $\langle iI, S_C\rangle$ in the normalizer of S_C.

Proof. Let $\psi_E = \Pi_C E |T\rangle^{\otimes t}$, with $\Pi_C = 2^{r-t} \sum_{g \in S_C} g$. Since L commutes with Π_C and $E^\dagger L g E = (-1)^{\langle E, Lg\rangle} Lg$, we have $\langle \psi_E | L | \psi_E \rangle = \langle T|^{\otimes t} E^\dagger L \Pi_C E |T\rangle^{\otimes t} = 2^{r-t} A_E(L)$. The operators $\Pi_C L \Pi_C$ span the operators on C, so the rank-one operators $|\psi_E\rangle\langle\psi_E|$ and $|\psi_0\rangle\langle\psi_0|$ are proportional, with ratio $A_E(I)/A_0(I) \geq 0$, exactly when the identities hold, and $\psi_E \in \mathbb{C}\,\psi_0$ exactly when they are proportional, including $\psi_E = 0$. Replacing L by Lg with $g \in S_C$ does not change $A_E(L)$. □

Each $A_E(L)$ has the form $a + b\sqrt{2}$ with dyadic rationals a and b, so each identity is a pair of identities between rationals. For $E = Z^e$ the sums are the signed weight enumerators with which Rall described distillation [36]. The test needs 4^r sums over the 2^{t-r} elements of S_C, in place of state vectors of dimension 2^t. We compared it with state vectors for all 7168 Pauli operators on 25 random codes with $t \leq 5$ and $r \leq 2$, without a disagreement.

Linear-programming bound. For an $[[n, k]]$ stabilizer code with stabilizer group S, let A_j and B_j be the numbers of elements of weight j of S and of its normalizer, up to phases, and let S_j be the number of Pauli operators of weight j in its shadow, which commute with the elements of even weight of S and anticommute with those of odd weight. These numbers satisfy [26, 27]

$$A_0 = 1, \qquad \textstyle\sum_j A_j = 2^{n-k}, \qquad B_j = 2^{k-n} \sum_i K_j(i) A_i, \qquad S_j = 2^{k-n} \sum_i (-1)^i K_j(i) A_i,$$
$$B_j \geq A_j \geq 0, \qquad S_j \geq 0 \qquad (0 \leq j \leq n),$$

where $K_j(i) = \sum_s (-1)^s 3^{j-s} \binom{i}{s} \binom{n-i}{j-s}$ is the coefficient of y^j in $(1+3y)^{n-i}(1-y)^i$, and distance d adds $B_j = A_j$ for $1 \leq j < d$. If no real A satisfies this system, there is no $[[n, k, d]]$ stabilizer code, nor one of smaller length, since appending a qubit in $|0\rangle$, with stabilizer Z, keeps the distance. For each entry of Table SI, the script

`scripts/cultivation_codes.py` (`--lp`) computes a Farkas certificate of infeasibility at length $n-1$ by an exact phase-one simplex method in rational arithmetic, and checks it in integer arithmetic. The system is satisfied by the weight distributions of the nine-input, eight-input and 15-input codes, and for $k=1$ and odd d its values coincide with Rains' bound $3d-4$.

TABLE SI. Lower bounds on the length n of a stabilizer code with k logical qubits and distance d from the linear-programming bound with shadow, certified in exact arithmetic, and the corresponding lower bounds on $n_{\min}$ of Ref. [17]. By Corollary 9, $n_{\min}$ is at least the larger of the two, for one T state ($k=1$) and for $|\mathrm{CCZ}\rangle$ ($k=3$).

$k=1$, d	6	7	8	9	10	11	12
$n \geq$	16	17	22	23	28	29	34
Ref. [17]	13	14	16	18	20	22	24
$k=3$, d	4	5	6	7	8		
$n \geq$	12	14	17	20	23		
Ref. [17]	11	12	13	15	17		

The construction. The same script builds the circuit of Theorem 8 for three codes, with every operation a noisy location, and computes its fault map. Pairs of faults are enumerated through their detector vectors, since a pair is undetected exactly when its two faults have the same D. With the eight-input CCZ code (8 T gates, 17 qubits, 97 two-qubit gates, one round of checks) and the nine-input T code (9, 16 and 128, one round), none of the 1860 and 2450 single faults is harmful, so both circuits have fault distance two, the order of their codes. With the 15-to-1 code and two rounds (15 T gates, 24 qubits, 458 two-qubit gates), none of the 8612 single faults and none of the 37,028,662 pairs is harmful; with one round (310 two-qubit gates), none of the 5874 single faults and 17,214,903 pairs is. The script also checks Lemma 3(ii) for these codes: all 27, 24 and 990 Pauli operators that act on fewer than d_C inputs collapse.

The published circuit. For the circuit of Proposition 11, all Z patterns of weights one and two on its 15 T gates collapse and 35 of weight three do not, so $d_C = 3$. Undetected single faults have $|E|_S \leq 1$. Of the undetected pairs of faults at distinct locations, 10,942, 12,338, 3887, 60 and 301 have $|E|_S = 0, \ldots, 4$ and are harmless, and 87 have $|E|_S = 5$, of which 48 are harmful. A Pauli of weight at least d_C modulo S_C is thus necessary for harm but not sufficient.

S3. THE SINGLE-QUBIT CHECK

The smallest circuit with a check prepares $|+\rangle$, applies the injection T gate, then $\mathrm{T}^\dagger$, projects onto $X=+1$ and applies T. The fault-free output is $|T\rangle$, since $|T\rangle$ is the $+1$ eigenstate of $TXT^\dagger$. In the skeleton the three T gates become ideal inputs a_1, a_2, a_3. The first two gadgets project the data qubit onto the same computational-basis state, and $S^\dagger$ contributes the phase $(-i)^b$, so the skeleton projects $a_1 a_2$ onto $|00\rangle + i|11\rangle$ and passes a_3 to the output; C has stabilizers Z_1Z_2 and X_1Y_2, and $\langle 00| + (-i)\langle 11|$ applied to $|T\rangle^{\otimes 2}$ gives $(1 - i\omega^2)/2 = 1$. Table SII lists every single fault. No fault triggers a detector of the skeleton; the check rejects Z-type errors through the collapse condition with zero, and projects X-type errors after the injection back onto $|T\rangle$, as in Fig. 1a. Faults after the last T gate are harmful, since nothing protects the output.

S4. COMPUTING THE FAULT MAP

The script `scripts/cultiv.py` computes D and E for a non-adaptive circuit given as a list of preparations, Clifford gates, T and $\mathrm{T}^\dagger$ gates, projections (with or without reset of the measured qubit), discards of qubits in known eigenstates, and identity operations that carry idle noise. It follows the Choi state of the skeleton on the physical qubits, the gadget inputs a_j and their references r_j, as in the proof of Theorem 2: the stabilizer generators are stored as binary vectors, and the sign of each generator as a linear form in the flips σ of all projections, which is all that the maps need, since constant signs only affect phases. A projection whose Pauli anticommutes with a generator replaces it, with the form σ_k; one whose Pauli is in the group adds the detector row $\sigma_k + \sum_i \ell_i$, where ℓ_i are the forms of the generators that multiply to it. For a fault, Pauli propagation through the skeleton gives the flips $s(F)$ and the output Pauli $P_{\mathrm{out}}(F)$. Then $D(F)$ evaluates the detector rows; the generators whose forms evaluate to one on $s(F)$

TABLE SII. Single faults of the single-qubit check. E is a representative modulo $\langle iI, S_C\rangle$ with $S_C = \langle Z_1Z_2, X_1Y_2\rangle$, on inputs $a_1a_2a_3$. The last column is the acceptance probability relative to the fault-free run.

location	fault	E	harmless	acceptance
after preparation of $\|+\rangle$	X	III	yes	1
	Y, Z	IZI	yes (rejected)	0
after the injection T	X	IYI	yes	1/2
	Y	IXI	yes	1/2
	Z	IZI	yes (rejected)	0
after $\mathrm{T}^\dagger$, before the projection	X	III	yes	1
	Y, Z	IZI	yes (rejected)	0
projection	sign flip	IZZ	yes (rejected)	0
after the last T	X, Y, Z	IIX, IIY, IIZ	no	1

change sign, which the destabilizers convert into a Pauli G on physical qubits and references; multiplying GP_{out} by the stabilizer with the same physical part, found by elimination, leaves a Pauli on the references, which is $E(F)$, reduced modulo S_C by a fixed echelon basis, so that the reduction is linear. When the circuit does not map onto its whole output space, as when the output is encoded and decoded, part $U(F)$ of the physical Pauli may remain unmatched; it is kept, and it vanishes whenever $D(F) = 0$ for the circuits of Theorem 2. The triple (D, U, E) of a configuration is the sum of those of its faults.

For $D(F) = 0$, harmlessness depends only on (U, E) and is decided by one simulation: a Pauli $E = X^uZ^v$ on the T inputs is the same as replacing T^k by T^{k-2} at the gates in u, which is what the gadget does with input $X|T\rangle$, and applying Z after the gates in v. The circuit so modified, followed by U, is simulated with state vectors and its output compared with the fault-free one. Pairs of faults are enumerated through the classes of their triples, so that each class is simulated once.

The script `scripts/cultiv_test.py` compares harmlessness from the fault map with direct simulation of the faulty circuit on 300 random circuits of two or three qubits with up to three T gates and two projections, for all single faults and random pairs, 42,367 configurations in all, without a disagreement. For the 65 circuits whose skeleton map has full row rank it also computes the skeleton map M_F densely, by feeding basis states to the gadgets, and checks for 9,271 configurations that $M_F = 0$ exactly when $D(F) \neq 0$ and that otherwise $U(F) = 0$ and $M_F \propto ME(F)$.

S5. EXACT EVALUATION AT SCALE

The circuits of Propositions 14–18 have up to 254 qubits and 77 T gates, beyond state vectors. We evaluate them with an independent implementation built on stim [20], in which the fault map of Theorem 2 is a detector error model and the collapse condition is decided by the Gauss sums of Theorem 5.

Skeleton and Choi state. Every T ($\mathrm{T}^\dagger$) gate on qubit q becomes CX_{q,a_j}, a measurement of a_j in the Z basis (and $S_q^\dagger$), with a_j half of a noiseless Bell pair with a reference r_j; noise channels stay where they were, and the gadgets, Bell pairs and final decoding are noiseless. A noiseless tableau run fixes the reference outcomes: every outcome that is random in the skeleton is fixed so that the circuit's own detectors read zero, and the remaining free outcomes are chosen. The state left on the references and the output qubit is the Choi state $|\Phi\rangle$ of $V\Pi_C$, with $t + 1$ stabilizer generators g_i.

Detectors and observables. The detectors are a basis of the record parities that are deterministic in the noiseless skeleton, found as the GF(2) null space of noiseless samples; the observables are the g_i, measured at the end, each completed by the record parity that makes it deterministic. One detector error model of the skeleton then gives every fault class with probability p_c, detector pattern D_c and observable flips o_c. For $D = 0$, the effect of the faults on the Choi state is a Pauli R on the references whose commutation pattern with the g_i is o; R is obtained from o by a fixed right inverse over GF(2), and $R = E(F)$ up to transposition.

Exact evaluation. Row reduction of the g_i writes $|\Phi\rangle = \sum_{x\in x_0+\mathrm{span}(u_1,\dots,u_k)} c(x)|x\rangle$ with $c(x_0) = 1$ and $c(x+u_i) = i^{\lambda_i}(-1)^{z_i\cdot x}c(x)$. For $R = X^aZ^b$ the output is

$$v_y = \sum_{x:\, x_{\mathrm{out}}=y} c(x)(-1)^{b\cdot x}\omega^{|x_{\mathrm{ref}}\oplus a|}, \tag{S4}$$

and the acceptance and bad weight of the fault class are $|v|^2/|v_0|^2$ and $|v|^2/|v_0|^2 - |\langle v_0, v\rangle|^2/|v_0|^4$. A Z pattern e acts through $\beta(e) = (e \cdot u_1, \ldots, e \cdot u_k) \in \mathbb{F}_2^k$ and the sign $(-1)^{e\cdot x_0}$, so one Walsh–Hadamard transform of length 2^k over the coordinates gives the output of every Z pattern. Definition 4 holds when exactly two values of β give a nonzero output, 0 and $\beta(\ell)$; then $\Sigma = \beta^{-1}(0)$, $\chi(e) = (-1)^{e\cdot x_0}$, and the conditions $w + b \in c\ell + \Sigma$ of Eq. (4) are the linear equations $\beta(w) = \beta(b) + c\beta(\ell)$. Every further class is evaluated with the Gauss sums of Proposition 6; the direct sum above, of cost 2^k, serves as a check. The affine dimension is $k = 5$ for the $d = 3$ circuit of Ref. [1], 15 for its $d = 5$ circuit, 11 for the one-round cat checks at $d = 5$, 21 for two rounds and 20 at $d = 7$.

Searches and rates. The fault distance is the least k for which some k fault classes have total detector pattern zero and a harmful total o. Detector patterns are compressed by a random GF(2)-linear 64-bit hash. Weight three uses a sorted table of single hashes and checks every hit on the exact patterns. Weight four groups pairs of classes by hash and evaluates every distinct observable mask of two pairs in the same group; a hash collision can only add masks, so the certificates that find no harmful mask are exact, and the harmful masks found are confirmed by a 4-set whose exact detector patterns cancel. Weight five splits each configuration canonically into its three smallest and two largest classes. The logical error at moderate p comes from sampling the detector error model with the exact acceptance and bad weight of each sampled class. For rare failures we use the decomposition $P_{\mathrm{bad}} = \prod_c (1-p_c) \sum_k A_k$ with $A_k = \sum_{|S|=k} \prod_{c\in S} r_c \, [D_S = 0] \, \mathrm{bad}(o_S)$ and $r_c = p_c/(1-p_c)$, and evaluate A_k by drawing $k-2$ classes with probability proportional to $\prod r_c$ and summing exactly over all completing pairs. On the $d = 3$ circuit this agrees with the sampled logical error within statistical error.

Validation. On the $d = 3$ circuit of Ref. [1] with real T gates, a state-vector simulation of all 2,444 single physical faults and 400 random pairs, each inserted explicitly and postselected on the reference outcomes, agrees with the algorithm in rejection for all 2,224 rejected configurations and in acceptance and bad weight to 3×10^{-16} for the 620 accepted ones; all 96 malignant pairs agree to the same precision and have acceptance and bad weight 1/4. With S in place of T, the sampled acceptance and logical error at $p = 3 \times 10^{-3}$ agree with direct stim sampling of the proxy circuit (acceptance 0.3251 in both, logical error $1.08(13) \times 10^{-5}$ against $1.02(12) \times 10^{-5}$). The Gauss sums agree with the direct sum on random sums of one to four fault classes: 335 classes of the $d = 3$ double check, 380 of the $d = 5$ protocol, 356 of the one-round verified cat at $d = 5$ and 60 of the unverified cat at $d = 7$, all to 5×10^{-16}; at $k = 20$ the direct sum takes 0.22 s per class and the Gauss sums 1 ms.

S6. THE NINE-T CIRCUIT

Table SIII lists the circuit of Proposition 10. Qubits 0 to 8 are prepared in $|+\rangle$ and receive a T gate each. The kth measurement prepares the ancilla in $|+\rangle$, applies controlled-P gates from the ancilla to the listed qubits in the listed order, with P the letter of g_k on that qubit, and measures the ancilla in the X basis, keeping the outcome equal to the sign of g_k. In the flagged measurement, a flag qubit in $|0\rangle$ is the target of CNOTs from the ancilla after the first and before the last controlled-Pauli gate, and is measured in the Z basis with outcome +1 kept. The ideal decoder measures $g_1, \ldots, g_8$ with their signs and unencodes. The circuit has 18 preparations, 9 T gates, 34 two-qubit gates and 9 noisy measurements, 70 locations in all.

TABLE SIII. The measurements of the nine-T circuit, in order: stabilizer of the nine-input factory code (qubits 0 to 8 from left to right), order of the controlled-Pauli gates, and flag.

k	g_k	order	flag
1	$-IYIXIXIIY$	8, 5, 3, 1	yes
2	$-IZZIIIIXX$	8, 2, 7, 1	
3	$-IIIIIZZXX$	6, 8, 7, 5	
4	$-YYXXIIIII$	2, 0, 3, 1	
5	$+IIIXXIIYY$	7, 4, 3, 8	
6	$+XIIZIZIXI$	5, 0, 3, 7	
7	$-XZIIZIIIX$	4, 0, 1, 8	
8	$-IIXXIIYYI$	6, 2, 3, 7	

Integer program. Let $\mathcal{W}$ be the 18 elements of weight four of the stabilizer group S of the factory code, with their signs, which span S; let $\langle \cdot, \cdot \rangle$ be the symplectic form; and let $\mathcal{C}$ be the 48 minimal linearly dependent subsets of $\mathcal{W}$. For $w \in \mathcal{W}$, an order π of its support and $f \in \{0, 1\}$, the measurement circuit (w, π, f) is that of Table SIII,

with a flag if $f = 1$. Let $\mathcal{R}(w, \pi, f)$ be the set of data Paulis R of the single faults in this circuit that raise no flag, satisfy $\phi = \langle R, w\rangle$ and do not collapse; by the proof of Proposition 10, these are the faults that are harmless only if R anticommutes with a stabilizer measured earlier. They are computed once, by Pauli propagation and the collapse test on nine T states; every Pauli that collapses has weight at most one modulo S. Binary variables s_w select the measured stabilizers, y_{vw} equals one if v is measured before w, and $z_{w\pi f}$ selects the order and the flag of the measurement of w:

$$
\begin{aligned}
\min_{s,y,z} \quad & \sum_{w\in\mathcal{W}} \sum_{\pi} z_{w\pi 1} \\
\text{s.t.} \quad & \textstyle\sum_{w\in\mathcal{W}} s_w = 8, \qquad \sum_{w\in C} s_w \le |C| - 1 \quad (C \in \mathcal{C}), \\
& \textstyle\sum_{\pi,f} z_{w\pi f} = s_w \quad (w \in \mathcal{W}), \\
& y_{vw} \le s_v, \quad y_{vw} \le s_w, \quad y_{vw} + y_{wv} \le 1, \quad y_{vw} + y_{wv} \ge s_v + s_w - 1 \quad (v \ne w), \\
& y_{uv} + y_{vw} + y_{wu} \le 2 \quad (u, v, w \text{ distinct}), \\
& z_{w\pi f} \le \textstyle\sum_{v\in\mathcal{W}:\, \langle R,v\rangle=1} y_{vw} \quad (R \in \mathcal{R}(w, \pi, f)), \\
& s_w,\ y_{vw},\ z_{w\pi f} \in \{0, 1\}.
\end{aligned}
$$

The first line selects eight independent stabilizers, which then generate S; the next two lines make y a linear order of the selected ones, since a tournament without directed triangles is transitive; the last line requires every dangerous Pauli to anticommute with a stabilizer measured before. The program has 1188 variables and 4777 constraints; Gurobi 10 finds a design with one flag in 46 s, which is optimal because the first measurement needs a flag. The script `scripts/cultivation_t9.py` builds the program (`--mip`), verifies the circuit, and writes `results/cultivation/t9_circuit.json`. Of the 627 single faults, 477 are detected, 13 rejected, 45 trivial and 92 absorbed (Fig. 1c), and 7146 of the 192,546 pairs are harmful.

Depolarizing noise. Under the uniform depolarizing model of Sec. S7 without idle noise, 6279 pairs of faults with nonzero probability are harmful, and the logical error per accepted run is $c_2 p^2 + O(p^3)$ with $c_2 = 53.1$; pairs of faults on the preparations and T gates, which act as faults of the factory's inputs, contribute 576 of the harmful pairs. For idle noise, we schedule every operation as early as possible, except that the ancilla and flag are prepared just before their first gate. The circuit then has 411 idle locations, 409 of them on the data, 132,968 pairs are harmful, and $c_2 = 1.9 \times 10^3$, mostly from pairs of idle faults on the data, which wait while the eight stabilizers are measured one after another.

S7. THE PUBLISHED CIRCUIT

Reconstruction. The repository of Ref. [28] defines the circuit of Proposition 11 as two stim chunks, the unitary injection (with one round of stabilizer measurements) and the distance-three double-cat check, whose boundary conditions are measurements of the color-code stabilizers and of the logical Y observable; S and S_DAG stand for T and $\mathrm{T}^\dagger$. We copied both chunks (`data/gsj_cultivation/`), joined them by the coordinates of their qubits, dropped the boundary measurements between them, kept the final stabilizer measurements as an ideal decoder, and replaced S and S_DAG by T and $\mathrm{T}^\dagger$. All measurements of the result are deterministic, both with T and with S, and it accepts with probability one without faults, so the decoder is admissible. With T gates, 4 of the 20 projections are not deterministic in the skeleton: the X measurement of the check (on a data qubit, which collects the logical parity through the cat state) and the three X-type stabilizer measurements of the decoder. These act through the code C, so the ideal decoder takes part in rejecting faults, as any decoder that postselects on all stabilizers does; we assume that the published state-vector simulations postselect in the same way, which the agreement of the coefficient below supports. Exchanging T and $\mathrm{T}^\dagger$ throughout gives the same 48 harmful pairs. The repository composes the chunks with its own compiler and re-layers the result; we did not run that code, so our idle noise follows the time steps of the chunk text.

Noise. With one location per operation, every Pauli at every preparation, gate and measurement and every flip of a measurement is a fault, 1397 in all on 137 operations. The uniform depolarizing model of Ref. [28] applies, with strength p, single-qubit depolarization after T (S) gates, two-qubit depolarization after CNOTs, a bit flip after a Z reset and a phase flip after an X reset, a flip of every measurement result, and single-qubit depolarization to every qubit that is idle in a time step; we apply the last only to qubits between preparation and measurement, since noise on other qubits is removed by the next reset. This gives 436 locations, 299 of them idle. The logical error per accepted run is $\epsilon = \sum_F w_F h_F / \sum_F w_F \operatorname{Tr}\Lambda(\rho_F)$, with h_F the probability that run F is accepted with a wrong output; the denominator is $1 + O(p)$, and the coefficient of p^2 in the numerator is a sum over the 96 harmful pairs (24 of two

CNOT faults, 24 of a CNOT and a T fault, 32 of an idle and a CNOT fault, 8 of an idle and a T fault, 8 of two idle faults) of the product of their probabilities divided by p^2, times the relative acceptance (1/4 for each) and the infidelity (1/2 or 1) of the output. It equals 32/75, the leading coefficient of the exact series of Wan and Zapirain [16], whose circuit was compiled independently and has 518 fault locations; the coefficient therefore does not depend on our layering of the idle noise. The S version has no harmful pair.

Comparison. Table SIV lists the state-vector simulations of Ref. [28] for this circuit, with T and with S gates at the same positions. A weighted least-squares fit of $(\epsilon_T - \epsilon_S)/p^2 = a + bp$ gives $a = 0.44 \pm 0.10$ for $p \le 3 \times 10^{-3}$, 0.45 ± 0.08 for $p \le 5 \times 10^{-3}$ and 0.42 ± 0.06 for all p, against $32/75 \approx 0.427$. The ratio ϵ_T/ϵ_S grows from 1.4 at $p = 10^{-2}$ to 2.5 at $p = 10^{-3}$, as a term of order p^2 in ϵ_T alone requires. The exact value of Ref. [16] at $p = 10^{-3}$ is $\epsilon_T = 8.98 \times 10^{-7}$; its series continues with $(2009039/4500)x^3$ in $x = p/(1-p)$, so the leading term is less than half of ϵ_T at this p.

TABLE SIV. Logical error per accepted run of the circuit of Proposition 11 under uniform depolarizing noise, from the state-vector simulations of Ref. [28] with T and with S gates (one standard deviation), and the acceptance probability with T gates.

p	ϵ_T	ϵ_S	acceptance	$(\epsilon_T - \epsilon_S)/p^2$
0.001	$(9.7 \pm 0.5) \times 10^{-7}$	$(3.9 \pm 0.3) \times 10^{-7}$	0.69	0.58 ± 0.06
0.002	$(5.86 \pm 0.15) \times 10^{-6}$	$(2.87 \pm 0.11) \times 10^{-6}$	0.47	0.75 ± 0.05
0.003	$(1.81 \pm 0.06) \times 10^{-5}$	$(1.03 \pm 0.04) \times 10^{-5}$	0.33	0.88 ± 0.08
0.005	$(8.75 \pm 0.28) \times 10^{-5}$	$(5.81 \pm 0.18) \times 10^{-5}$	0.15	1.18 ± 0.13
0.007	$(2.60 \pm 0.08) \times 10^{-4}$	$(1.90 \pm 0.06) \times 10^{-4}$	0.074	1.42 ± 0.21
0.010	$(9.1 \pm 0.3) \times 10^{-4}$	$(6.7 \pm 0.2) \times 10^{-4}$	0.025	2.4 ± 0.4

Flags. A flag on qubit q over the window $[t_1, t_2)$ of operations of the check is a qubit prepared in $|0\rangle$, the target of a CNOT from q before operations t_1 and t_2, and measured in the Z basis with outcome $+1$ kept. It is a detector when no operation in the window changes the Z value of q (no CNOT targets q, and q is neither measured nor reset), and it detects a fault F exactly when the X part of the Pauli frame of F on q differs at t_1 and t_2; write $\chi_K(F) \in \{0,1\}$ for this, for each of the 5590 such candidates K. A pair (F_1, F_2) is detected by K when $\chi_K(F_1) \neq \chi_K(F_2)$, so the fewest flags that detect every harmful pair solve

$$\min_x \sum_K x_K \quad \text{s.t.} \quad \sum_{K:\,\chi_K(F_1)\neq\chi_K(F_2)} x_K \ge 1 \text{ for every harmful pair } (F_1,F_2), \qquad x_K \in \{0,1\}.$$

The program is infeasible: for two pairs, in which the X errors arise on targets of CNOTs, no candidate detects either fault. Covering the other 46 pairs takes eight flags. Chan *et al.* restore fault distance three with two flag qubits and four additional layers of CNOTs [15], so flags outside this family suffice. The flags themselves would add faults, which this model ignores; the infeasibility does not depend on them. The script `scripts/cultivation_gsj.py` carries out all computations of this section and writes `results/cultivation/gsj_d3.json`.

The circuits of Ref. [1]. We use the published `inject[unitary]+cultivate` circuits at $d_1 = 3$ (15 qubits, one double check) and $d_1 = 5$ (42 qubits, double checks at distance three and five, with growth in between), with their uniform noise including idling, read with real T gates. For the SI1000 model we keep the idle sets of the circuits and set the two-qubit gate noise to p, single-qubit gates (including T) and idling to $p/10$, resets to $2p$, measurements to $5p$ and idling during measurement or reset layers to $2p$ [32]. At $d_1 = 5$ the $\mathrm{T}/\mathrm{T}^\dagger$ pattern on the 19 data qubits places $\mathrm{T}^\dagger$ on 18 qubits and T on the qubit shared by the three hexagons in the first layer, and the reverse in the second; the complementary pattern satisfies the same face conditions but checks the conjugate state. Of the branches of the random growth outcomes, the default reference branch has maximal weight, that is, no X frame before the $d = 5$ T layers. At $d_1 = 3$ under uniform noise the sampled logical errors agree with the state-vector data of Table SIV, for instance $8.94(33) \times 10^{-7}$ at $p = 10^{-3}$ against $(9.7 \pm 0.5) \times 10^{-7}$; Table SV lists the other cases.

Malignant configurations. At $d = 3$ the nine malignant fault classes of weight two consist of an X or Y fault on a fold qubit and an X or Y fault on a data qubit at the final T layer; immediately before that layer they leave X on two or three data qubits. At $d = 5$ the nine classes of weight three consist of two hooks and one data fault, and leave X on four or five data qubits; their acceptance is 2^{-4} and their bad weight 2^{-4} or 2^{-5}, as Lemma 12 predicts for Y and X.

TABLE SV. Logical error per accepted output of the circuits of Ref. [1] with real T gates and with the S proxy. Real T and proxy have the same acceptance. At $d = 5$ the proxy error is of order p^5; the last column is the GPU simulation of Ref. [9], whose circuit is a modified version of the same protocol.

circuit	noise, p	acceptance	real T	proxy	Ref. [9]
$d = 3$	SI1000, 10^{-3}	0.723	$8.00(21) \times 10^{-7}$	$2.84(22) \times 10^{-7}$	
	SI1000, 5×10^{-4}	0.850	$1.52(4) \times 10^{-7}$	$3.35(35) \times 10^{-8}$	
$d = 5$	uniform, 2×10^{-3}	0.021	$3.6(7) \times 10^{-8}$		3.4×10^{-8}
	uniform, 10^{-3}	0.144	$3.26(21) \times 10^{-9}$		4.6×10^{-9}
	uniform, 5×10^{-4}	0.379	$2.72(12) \times 10^{-10}$		1.6×10^{-10}
	SI1000, 10^{-3}	0.155	$2.2(6) \times 10^{-9}$		
	SI1000, 5×10^{-4}	0.394	$1.09(8) \times 10^{-10}$		

S8. FURTHER CIRCUITS

Flags. A Z-type flag on qubit q couples $\mathrm{CX}_{q,f}$ at two layer boundaries and measures f in the Z basis; it flips when the X part of the Pauli frame on q differs at the two boundaries, and it is non-disturbing when Z_q at the two boundaries has a deterministic product in the skeleton, which we test by a tableau simulation. X-type flags are defined dually, and two-qubit flags couple to Z_aZ_b. The integer program chooses flags that flip for every physical representative of every malignant configuration, with constraints that involve faults of a flag or of an inserted layer imposed only when that flag or layer is chosen, and every design is re-certified exactly. For single-qubit flags with both couplings between the first T layer and the unpairing layer of the $d = 5$ double check, some representatives of the malignant configurations flip no candidate flag; they remain malignant in every flagged circuit, which proves Proposition 15. With two-qubit flags, the program finds two flags, Z_3Z_7 and $Z_{32}Z_{33}$ (qubit labels of the published circuit), coupled over the whole fold and unfold, which give fault distance four. Their logical error is $1.07(15) \times 10^{-10}$ against $2.72(12) \times 10^{-10}$ for the unflagged circuit at uniform $p = 5 \times 10^{-4}$ (acceptance 0.345 against 0.379), and $1.00(28) \times 10^{-9}$ and $3.1(6) \times 10^{-11}$ against $2.17(56) \times 10^{-9}$ and $1.09(8) \times 10^{-10}$ in SI1000 at $p = 10^{-3}$ and 5×10^{-4}, at acceptance 0.148 and 0.385 instead of 0.155 and 0.394. The remaining malignant configurations of weight four begin with X faults on two data qubits between the last stabilizer round and the first T layer, which no flag inside the check can see.

Verified-cat checks. The circuits of Proposition 16 at $d = 5$ and $d = 7$ start from an ideal encoded state, the injected state on one qubit and a CNOT encoder, followed by a flagged stabilizer round, one check round (or two, separated by a flagged stabilizer round) and a flagged stabilizer round, with uniform noise of strength p after every gate, on resets and measurements and on idle live qubits, and a noiseless final readout. The cat on n qubits is prepared by a CNOT tree from one qubit in $|+\rangle$ and verified by the $n-1$ parities $Z_{c_q}Z_{c_{q+1}}$, each with its own ancilla; the check outcome is the parity of the n X measurements of the cat. The injected state, $\mathrm{T}|+\rangle$ at $d = 5$ and $\mathrm{T}^\dagger|+\rangle$ at $d = 7$, is matched to the global parity of the $\mathrm{T}/\mathrm{T}^\dagger$ pattern. At $d = 5$, the certificate of fault distance five of one round evaluates exactly the 56,319 distinct observable patterns of undetected configurations of four faults and finds none harmful; the enumeration at weight five finds 754,515 malignant configurations with total weight 3.452×10^{-12} at $p = 10^{-3}$. Without verification, 99 malignant configurations of weight three have total weight 5.4×10^{-10}. For two rounds the weight-four certificate evaluates 3,220,729 distinct errors. At $d = 7$ ($t = 75$, $k = 20$), the weight-four certificate of the verified cat evaluates 464,480 distinct errors and finds none harmful; the unverified cat has no malignant configuration of weight three and 38 distinct harmful errors at weight four, each realized by four fault classes whose exact detector patterns cancel, with acceptance 2^{-5}, 2^{-6} or 2^{-7} relative to the fault-free run. At uniform $p = 10^{-3}$ the acceptance at $d = 5$ is 0.11 with verification and 0.23 without, and in SI1000 at $p = 5 \times 10^{-4}$ it is 0.17 and 0.30. The $d = 3$ verified-cat circuit applies T to all seven data qubits in each layer, with $\mathrm{T}^\dagger|+\rangle$ injected on one qubit before a non-fault-tolerant encoder, two check rounds and three flagged stabilizer rounds; it has fault distance three with real T gates, equal to its proxy distance, with 2,405 malignant configurations of weight three. With $\mathrm{T}|+\rangle$ injected instead, its checks project, Definition 4 fails with four accepted Z classes, and the fault distance is two.

GHZ-fold checks at $d = 3$, 5 and 7. The circuits of Proposition 14 have the surroundings of the cat checks and one check round built as the double check of Ref. [1]: n partners prepared in $|+\rangle$, the $\mathrm{U}^\dagger$ layer, CX from each partner to its data qubit, a fold CX(parent, child) over a spanning tree whose nodes are one representative (data qubit or partner) of each pair, leaves first, an X measurement and reset of the root, the fold in reverse, the pairing CNOTs again, the U layer and X measurements of the partners. The root and the product of the partners read the check

twice. The trees are spanning trees of the graph of data qubits that share a face, grown breadth-first from a central qubit or in random order, either unrestricted (depth one to three) or with at most two children per node (depth five to seven, as for the nearest-neighbour fold of Ref. [1]), with data representatives or random ones. Detectors are all record parities that are deterministic in the noiseless proxy, and the injected state, $\mathrm{T}^\dagger|+\rangle$ at $d = 3$ and 7 and $\mathrm{T}|+\rangle$ at $d = 5$, is the one for which Definition 4 holds. At $d = 3$ the three trees have 6, 12 and 12 malignant pairs, with leading terms $0.089p^2$, $0.57p^2$ and $1.0p^2$; at $d = 5$, 68, 85 and 3 malignant triples, with $0.43p^3$, $0.69p^3$ and $0.076p^3$, the last as few as for the best fold trees of the published circuit. At $d = 7$ ($t = 75$, $k = 20$) no tree has a malignant configuration of three faults, and the six trees have 15, 144, 80, 523, 362 and 352 distinct harmful errors at weight four, each realized by four fault classes whose exact detector patterns cancel; each certificate takes 8 to 17 minutes. The proxy distances follow from the stim detector error model of the proxy circuit with the same search.

Leading terms and acceptance at $d = 7$. The leading terms of Table I at $d = 7$ come from the exact enumeration of all malignant configurations of four faults: for the six fold trees of Proposition 14 (central, central binary, binary 1, binary 2, binary 3, random 1) there are 15, 256, 111, 957, 501 and 517 of them, with total weights 0.011, 0.24, 0.062, 15, 4.5 and 5.4 times p^4, and 120 with $0.093\,p^4$ for the unverified cat. On the flagged double check at $d = 5$ the same enumeration gives 6,824 configurations and $1.16 \times 10^3 p^4$, as in Table I. The acceptances of Table SVI come from sampling the detector error model with the exact acceptance of each sampled class (5×10^6 samples).

TABLE SVI. Acceptance of the checks of Table I built in our surroundings (one check round between flagged stabilizer rounds), under uniform noise; GHZ fold: central binary tree.

check	$d = 5$, 10^{-3}	$d = 5$, 5×10^{-4}	$d = 7$, 10^{-3}	$d = 7$, 5×10^{-4}
GHZ fold	0.050	0.222	0.0005	0.020
cat, not verified	0.226	0.474	0.015	0.120
cat, verified	0.110	0.330	0.0009	0.029

Nearest-neighbour checks. The checks of Proposition 17 start from the same ideal encoded state as the cat checks above, apply the $\mathrm{U}^\dagger$ layer, prepare the cat on the path, verify it, couple each data qubit once to its path site by a CNOT, un-grow the cat, measure the root in the X basis and the other path qubits in the Z basis, and apply the U layer; the final readout is noiseless, and there are no stabilizer rounds. Every two-qubit gate of the check acts on grid neighbours, which the construction asserts; a CNOT counts as one two-qubit gate (on CZ hardware it carries two Hadamards of noise $p/10$ each in SI1000). The path is grown from its middle site, one CNOT per arm per layer. In the ring ($d = 3$) the path has nine sites, seven next to data and two corner bridges, and the ancilla next to both ends measures $Z_a Z_b$; one round takes 15 layers. The $30\,p^4$ term of Proposition 17 is the sum over all 9,340 malignant configurations of four faults. Without un-growing, with the cat read out by nine X measurements, the leading term is $16\,p^3$ in SI1000 ($30\,p^3$ with two check rounds), and $6.9\,p^3$ when the end parity is measured twice. In the ladder ($d = 5$) the 19 path sites are the top row $(x, 0)$, $x = 0, \ldots, 7$, the turn $(8, 0), (8, 1), (8, 2)$ and the bottom row $(x, 2)$, $x = 7, \ldots, 0$, each next to one data qubit, with the root at $(8, 1)$; the eight middle sites $(x, 1)$ measure the eight rungs and, in pairs, the seven diagonals, the odd and even ones in two parallel layers, and each two-ancilla gadget uncomputes its second ancilla. One round takes 34 layers (24 without un-growing). The weight-four certificate evaluates 15,775 distinct errors in 7 s; the enumeration at weight five finds 85,464 malignant configurations (92,216 without un-growing, for which a second check round also keeps fault distance five, with 273,827 distinct errors at weight four). Without the diagonals the ladder has fault distance four, with nine malignant configurations of weight four. The $d = 7$ ladder has 37 path sites ($17 + 3 + 17$), 17 middle ancillas and 91 qubits, injects $\mathrm{T}^\dagger|+\rangle$, and its weight-four certificate evaluates 195,829 distinct errors in under three minutes; with the un-grown cat, the enumeration of all configurations of five faults whose detector patterns cancel finds no harmful one, in a quarter of an hour on 14 cores.

The complete nearest-neighbour protocol. The complete protocol of the main text uses the ring layout with every operation noisy. The data are reset, $\mathrm{T}^\dagger$ is applied to the carrier and H to it and to the pivots of the encoder, and each data-data CNOT fan-out of the encoder is relayed: the path segment covering the control and its targets is reset, the control is copied onto its path site, spread along the segment, applied to the targets from their own path sites, unspread and uncopied, and the segment is measured (deterministically zero). The three Z and the three X stabilizers are then measured, each with a verified path cat coupled to the four data of the face (by CNOTs conjugated with Hadamards on the data for Z), followed by three checks with the cat read out by X measurements. The circuit has 17 qubits and 107 layers, every two-qubit gate between grid neighbours, real-T and proxy fault distance three, and 10,573 malignant configurations of weight three with leading term $2.8 \times 10^3 p^3$ under uniform noise and $7.8 \times 10^3 p^3$ in SI1000, at acceptance 0.11 in SI1000 at $p = 10^{-3}$. With two checks the leading term in SI1000 is $4.8 \times 10^4 p^3$, and

with un-grown cats the deeper circuit has $6.1 \times 10^4 p^3$. Measuring the Z stabilizers by folding their parity along the path instead leaves weight-two Z hooks of the encoder undetected, and the fault distance is two. In the alternative that prepares $|0\rangle_L$ by measuring the X stabilizers with the path cat and lets the first check project, a Z error before the first check imitates a Z_L frame and an X error before it passes all T layers coherently, so the X and Z stabilizers must be measured again after the first check; the schedule X stabilizers, check, Z and X stabilizers, check, check has fault distance three and $7.1 \times 10^4 p^3$ in SI1000 at acceptance 0.05.

Google's circuits. Ref. [2] publishes its circuits as stim files in which every non-Clifford rotation is an identity tagged `T_gate`, `T_dagger_gate` or `injection`, and the physical Hadamards that turn the Y rotations into virtual Z rotations are tagged `conjugate`. We replace the tagged identities by H_{YZ}, T and $\mathrm{T}^\dagger$, and write the injection $R_y(\pi/4)|0\rangle$ as $\sqrt{Y}^\dagger$, $\mathrm{T}^{\pm 1}$ and H_{YZ}, the same state, with the sign for which the fault-free circuit accepts with certainty; the other sign halves the acceptance. In the circuit of their Fig. 3 (14 qubits: seven data qubits, injection, three error-correction cycles and three checks, the last of which is the kickback measurement), we cut after the first, second or third check, or after the following cycle, and read out the six stabilizers and Z_L noiselessly, as for the circuits of Ref. [1]; the detectors are all deterministic record parities, so every run with a detection event is discarded, as in the experiment. SI1000 noise is applied layer by layer with echo pulses as single-qubit gates, and layers that hold only T gates take no time. Every cut has real-T fault distance two with 20 malignant pairs and proxy fault distance three, with leading terms $0.44\,p^2$ to $2.2\,p^2$; making the injection and the first cycle noiseless leaves the 20 pairs and their weights unchanged. The single check of their Fig. 2, without cycles, has fault distance two with real T gates and with the proxy. The logical errors of Fig. 4 come from 1.2×10^9 to 1.1×10^{10} samples per point. Ref. [2] reports a logical error of $1\text{–}2 \times 10^{-4}$ at 8% retention for the kickback circuit with its own measurement, which includes the error of the kickback readout and noise outside the Pauli model; our numbers are those of the cultivated state before that readout.

Code-relative fault tolerance. Table SVII lists, for the triangular color codes, the fraction of X patterns of each weight on the data that lie inside the support of a minimum-weight logical X operator; by Corollary 13, those that contain no Z stabilizer may not be the result of fewer undetected faults than their weight in a check of full real-T fault distance.

TABLE SVII. Fraction of the X patterns of weight w on the data of the triangular color codes that lie inside the support of a minimum-weight logical X operator.

code (minimum-weight X logicals)	$w=2$	3	4	5	6
$[[7,1,3]]$ (7)	1				
$[[19,1,5]]$ (36)	0.895	0.291	0.046		
$[[37,1,7]]$ (140)	0.811	0.278	0.052	0.0060	0.0004

S9. FURTHER NUMERICAL CHECKS

The script `scripts/cultivation_check.py` checks Theorem 2 by an independent state-vector implementation. For each circuit it compares, for all single faults and up to 800 pairs of faults, the faulty output computed with physical T gates with the output of the skeleton fed with ideal T states; whether the skeleton map is zero or a multiple of ME for some Pauli E on the T inputs; harmlessness computed directly with the collapse condition; the claim $E = Z_j$ for Z faults at T gates; and $E(F_1 + F_2) = E(F_1)E(F_2)$ modulo $\langle iI, S_C\rangle$. For the targets $\mathrm{CS}|{+}{+}\rangle$ (three T gates and a ZZZ check) and $|\mathrm{CCZ}\rangle$ (the standard seven-T circuit), it also checks Corollary 20. We ran 12 circuits, the two above and ten random circuits on two qubits with two to four T gates and one or two projections. The gadget identity holds for all 10,542 configurations tested. In the six circuits whose skeleton map has full row rank, as Theorem 2 assumes, the structural statement and the harmlessness criterion hold for all 5,316 configurations, $E = Z_j$ holds for all 24 T gates, and multiplicativity for all 3,726 pairs tested; Corollary 20 holds for all 1,832 configurations of the two rational circuits. An independent dense simulation of the circuit of Proposition 11, written without shared code, confirms the harmful pairs that we checked, with relative acceptance 1/4 and infidelity 1/2 or 1.

S10. OPEN PROBLEMS

Space-time cost. The circuit of Proposition 10 minimizes T gates. The least number of qubits, two-qubit gates or time steps at fault distance two, and how many T gates fault distance three needs (between ten and fifteen, by Corollary 9), are open. By Theorem 8, the second is the distillation question of Ref. [17] at order three. Integer programs over the fault map, as in Sec. S6, apply to any family of circuits in which the fault map is linear in the design choices, such as orders of gates and positions of flags.

Larger circuits. Theorem 5 decides the collapse condition in polynomial time for circuits with deterministic checks, which is how the circuits of Propositions 14–18 were certified; the cost of a certificate is now the search over fault configurations, whose detector patterns must cancel. At $d = 7$ the weight-four searches take minutes and weight five a quarter of an hour; the nearest-neighbour ladder at $d = 7$ has no malignant configuration of five faults, and a search at weight six would decide whether it keeps the full distance.

Analytic bounds. The witnesses of Ref. [17] feed a protocol a state whose fault patterns are all harmless. With coherent faults, the analogous argument would feed the skeleton states of the form $E|T\rangle^{\otimes t}$ superposed over tolerated E.

Rates. Proposition 1 discards the acceptance probability, so bounds on expected cost per accepted state need a different argument.